

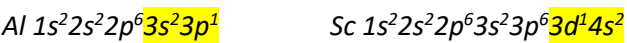
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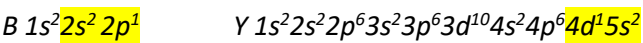
Periodic law is formulated as follows: *the properties of elements and their compounds vary periodically with the nuclear charge of the atoms of the elements.*

	1	2	3	4	5	6	7	8	
1	H ¹								He
2	Li ³	Be ⁴	⁵ B	⁶ C	⁷ N	⁸ O	⁹ F		Ne
3	Na ¹¹	Mg ¹²	¹³ Al	¹⁴ Si	¹⁵ P	¹⁶ S	¹⁷ Cl		Ar
4 	K ¹⁹	Ca ²⁰	Sc ²¹	Ti ²²	V ²³	Cr ²⁴	Mn ²⁵	Fe ²⁶ Co Ni	
	²⁹ Cu	³⁰ Zn	³¹ Ga	³² Ge	³³ As	³⁴ Se	³⁵ Br		Kr
5 	Rb ³⁷	Sr ³⁸	Y ³⁹	Zr ⁴⁰	Nb ⁴¹	Mo ⁴²	Tc ⁴³	Ru ⁴⁴ Rh Pd	
	⁴⁷ Ag	⁴⁸ Cd	⁴⁹ In	⁵⁰ Sn	⁵¹ Sb	⁵² Te	⁵³ I		Xe
6 	Cs ⁵⁵	Ba ⁵⁶	La* ⁵⁷	Hf ⁷²	Ta ⁷³	W ⁷⁴	Re ⁷⁵	Os ⁷⁶ Ir Pt	
	⁷⁹ Au	⁸⁰ Hg	⁸¹ Tl	⁸² Pb	⁸³ Bi	⁸⁴ Po	⁸⁵ At		Rn
7	Fr ⁸⁷	Ra ⁸⁸	Ac**	Ku ¹⁰⁴					
*	Ce ⁵⁸								Lu
**	Tb ⁹⁰								Lr

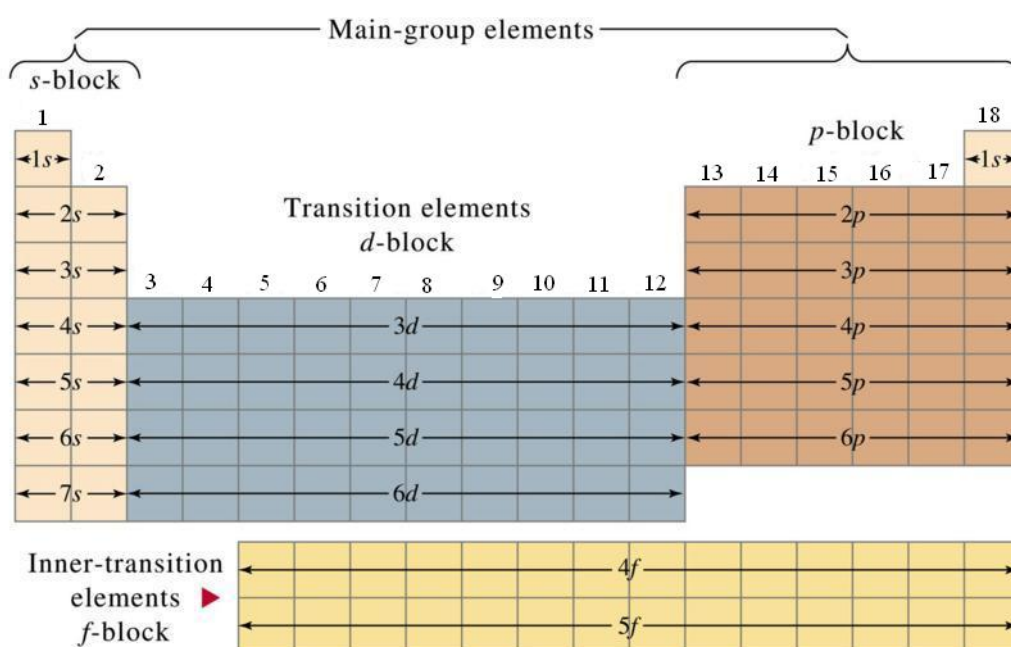
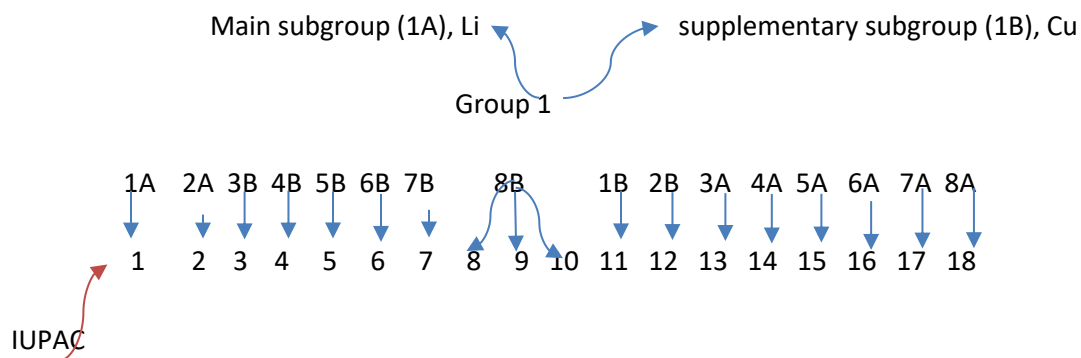
Same group because their valence orbitals accommodate the same number of electrons.



Same subgroup because their valence electrons are in orbitals with similar configurations.



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General trends in Main group chemistry

The main group elements are characterized by the completion of their electron configurations using *s* and *p* electrons. These elements cover the full range of classifications, from the most metallic to the most nonmetallic, with elements of intermediate properties, the semi-metals (metalloids), in between. On the far left, the alkali metals and alkaline earths exhibit the expected metallic characteristics of luster, high ability to conduct heat and electricity, and malleability. The distinction between metals and nonmetals is best made on their electrical conductivity. Alkali metals have low resistivity (high abilities to conduct), nonmetals have high resistivity. These dramatic differences in conductivity between metals and nonmetals can be explained on the basis of the bonding and structure of crystals of these elements. Metals contain loosely bound valence electrons that are relatively free to move and thereby conduct

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current, nonmetals contain much more localized lone electron pairs and covalently bound pairs, which are much less free.

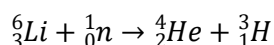
Elements along a rough diagonal from **boron** to **polonium** are intermediate in behavior, in some cases having both metallic and nonmetallic allotropes (elemental forms); these elements are designated **metalloids** or **semimetals**. Some, such as silicon and germanium, are capable of having their conductivity finely tuned by the addition of small amounts of impurities and consequently of enormous importance in the manufacture of semiconductors in the computer industry.

Electronegativity* also provides a guide to the chemical behavior of main group elements. **ability of an atom of a given element to pull to itself common electron pair.* The extremely high electronegativity of the nonmetal fluorine is evident, with a steady decline in electronegativity toward the left and bottom of the periodic table. The semimetals form a diagonal of intermediate electronegativity.

Ionization energies of the main group elements exhibit rather similar trends to electronegativity. While there is a general increase in ionization energy toward the upper – right corner of the periodic table, the change is irregular but has a tendency of growing.

Hydrogen H and its compounds

Hydrogen (Hydrogenium – water producing) has three isotopes: **protium** (^1H), **deuterium** ^2H or **D** and **tritium** ^3H or **T**. Tritium is produced in nuclear reactors by bombardment of ^6Li nuclei with neutrons:



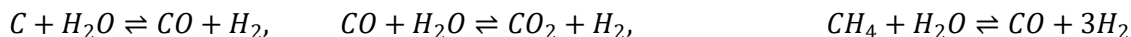
Protium and deuterium have stable nuclei; tritium undergoes beta decay, $^3_1\text{H} \rightarrow ^3_2\text{He} + e^-$

Tritium is used as a tracer, especially in metallurgy, in studying the movement of ground waters, and to study the absorption of hydrogen by metals and the adsorption of hydrogen on metal surfaces.

Deuterium is used to produce perdeuterobenzene, C_6D_6 , for use as an NMR solvent.

Practically, all of hydrogen is in a bonded state in many minerals, fuel, coal, animal and plant organisms, but its **most common compound** is **water**.

Industrial source of hydrogen is water and partly hydrocarbons:



In the laboratory hydrogen is obtained by reacting dilute solution of an acid with an active metal (usually, $\text{H}_2\text{SO}_4 + \text{Zn}$).

Other redox¹ reactions with participation of water: $2\text{Na} + 2\text{H}_2\text{O} = 2\text{NaOH} + \text{H}_2\uparrow$,

$\text{CaH}_2 + 2\text{H}_2\text{O} = \text{Ca}(\text{OH})_2 + 2\text{H}_2\uparrow$.

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¹**Oxidation number (state)** of an element in a compound is the № of electrons that have passed from one atom of a given element to other atoms (positive oxidation) OR

to one atom of a given element from other atoms (negative oxidation).

Oxidation state of oxygen in its compounds is, -2, except in peroxides (-1) and in OF₂ (+2).

Oxidation – losing of electrons by an atom attended by an increase in its oxidation number.

Reduction – gaining of electrons by an atom attended by a decrease in its oxidation number.

A substance containing an element that undergoes oxidation is called **reducing agent**.

A substance containing an element that undergoes reduction is called **oxidizing agent**.

$4\text{Al} + 3\text{O}_2 \rightarrow 2\text{Al}_2\text{O}_3$ here, Al is reducing agent, O₂ is oxidizing agent.

Intermolecular redox reactions – $\text{AB} + \text{CD} \rightarrow \text{EF} + \text{GH}$

Intramolecular redox reactions – (decomposition) – *different elements of a compound – one – oxidizer, the other – reducer*; $\text{KClO}_3 \rightarrow \text{KCl} + \text{O}_2$

Auto oxidation – auto reduction (disproportionation) redox reactions: *atoms of a particular element – one – oxidizer, the other – reducer*; $\text{Cl}_2 + \text{H}_2\text{O} = \text{HCl} + \text{HOCl}$ $4\text{H}_3\text{PO}_3 = 3\text{H}_3\text{PO}_4 + \text{PH}_3$

Physical properties: in normal conditions, hydrogen is the lightest gas, colourless, odourless, tasteless, poorly dissolves in water, a good conductor of heat (7 times better conductor than air).

Chemical properties: being placed between nonmetals and metals in terms of electronegativity, hydrogen acts as a reducer in its reactions with nonmetals and O – compounds: $2\text{H}_2 + \text{O}_2 = 2\text{H}_2\text{O}$, $\text{CuO} + \text{H}_2 = \text{Cu} + \text{H}_2\text{O}$.

With very active metals, hydrogen interacts directly as an oxidizer, changing into H⁻ anion (hydride ion) (KH, CaH₂,....). These crystal compounds are salts which fully dissociate when they interact with water to give H₂ and bases.

Practically, all elements form compounds with hydrogen – which can be divided into three main groups: **covalent**, **ionic** and **metallic**. Atoms of all elements from group IVA to VIIA form covalent bonds with hydrogen. Atoms of metals of groups IA and IIA (except beryllium and magnesium) form ionic bonds with hydrogen. All groups **B** elements (*d* – elements) form with hydrogen, metallic hydrides.

Hydrogen is a unique element in many ways and it is difficult to fit it into any particular group of the periodic table.

Hydrogen is often included in the first main subgroup because, like the alkali metals, it is an **s** – element; however, notwithstanding the characteristics that they have in common (similar spectra, formation of

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the univalent positive ion R^+ , reducing properties, mutual replacement of metals and hydrogen), there are essential differences between metals and hydrogen: the proton is incomparably smaller than the cations of the alkali metals and it always penetrates deeply into the electron shells of the atom with which it is united; the ionization energy of hydrogen is almost three times greater than the approximately equal first ionization energies of the alkali metals; the behavior of hydrogen is similar to that of metals only in aqueous solutions, it acts as a reducing agent only at high temperatures. On the other hand, hydrogen in many of its properties resembles the halogens. That is why it is more appropriate to include hydrogen in the fluorine subgroup. The other resemblance between hydrogen and the halogens: the gaseous state of hydrogen, the fact that its molecule contains two atoms, the ease with which hydrogen is replaced by halogens in organic compounds, the closeness of the energies of decomposition of H_2 and Hal_2 molecules, the comparability of the ionization potential of hydrogen and the first ionization potentials of the halogens. Of course, it should be kept in mind that hydrogen differs from the halogens (because the latter, being p – elements, form compounds in which the oxidation number is higher than unity).

However, the properties of hydrogen bear a greater resemblance to those of the halogens than to the properties of metals.

Water, its properties

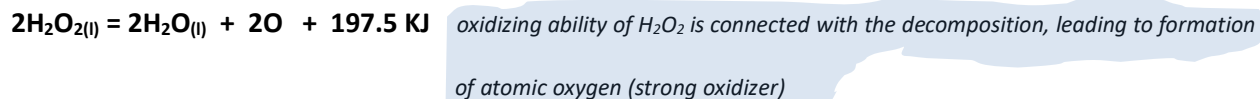
Three isotopes of hydrogen (1H , D and T) and three isotopes of oxygen (^{16}O , ^{17}O and ^{18}O) in different combinations can form 18 isotopically different types of water with molecular mass from 18 to 24 ($T_2^{18}O$).

Earth (ground water) contains light water, heavy water by oxygen and heavy water by hydrogen. Usually for heavy water (by hydrogen) we have – D_2O .

In heavy water, substances dissolve poorly, solutions conduct current poorly. It is hygroscopic (*it avidly absorbs vapour from the atmosphere*). In this water, fish will die, microbes will not survive.

Hydrogen peroxide H_2O_2

H_2O_2 is a colourless viscous (syrup – like) liquid. It is a very unstable substance, able to explode and disintegrate into water and atomic oxygen, releasing a lot of heat:



Aqueous solutions of H_2O_2 are more stable; in a cold place they can be stored for a longer period.

Perhydrol – solution on sale – contains 30% H_2O_2 . In it and in more concentrated solutions of H_2O_2 there is a stabilizing additive. Catalysts increase the disintegration of H_2O_2 . E.g., if in a solution of H_2O_2 , a small amount of MnO_2 is added, then there is a turbulent reaction and release of oxygen. Catalysts can be copper, iron, manganese and ions of these metals.

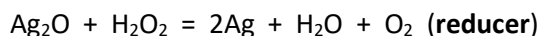
H_2O_2 reacts directly with some bases to produce salt: $Ba(OH)_2 + H_2O_2 = BaO_2 \downarrow + 2H_2O$

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Salts of H_2O_2 are called **peroxides**.

H_2O_2 display both **oxidizing** and **reducing** properties. This is because the oxidation state of oxygen in H_2O_2 lies between that of oxygen in H_2O , (-2), and oxygen in O_2 , (0). Examples:

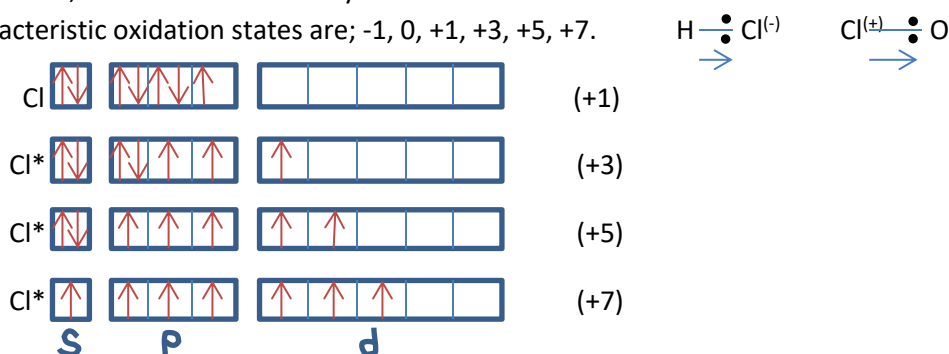


The use or application of H_2O_2 is connected to its oxidizing capability and the harmless product of its reduction (H_2O). It is used for bleaching clothe and fur, it is used in medicine as disinfectant (3% solution), for preserving food, in agriculture as seed disinfectant, for production of several organic compounds, polymers, also in rocket technology (because it is a strong oxidizer).

Group VIIA elements

Atoms of halogens that form group VIIA have electron configuration ns^2np^5 . All halogens are active nonmetals, oxidizers. Their activity reduces with increase in radius in the series $\text{F} > \text{Cl} > \text{Br} > \text{I} > \text{At}$.

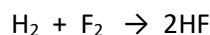
Characteristic oxidation states are; -1, 0, +1, +3, +5, +7.



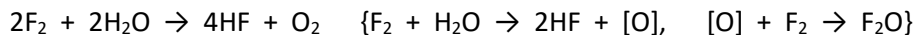
However, fluorine has only oxidation states; -1, 0. *Fluorine cannot be oxidized by any other substance.*

Fluorine and its compounds

Fluorine is the **most active nonmetal, strongest oxidizer**. It is a greenish yellow gas with a sharp (pungent) smell. Fluorine interacts with hydrogen with an explosion in darkness and at low temperature:



Fluorine cannot be dissolved in water since it oxidizes water (it energetically decomposes water):



Fluorine forms two oxides: OF_2 and O_2F_2 (*difluoride dioxygen*).

In air, fluorine burns many metals and nonmetals with the formation of the corresponding fluorides. All metals (except platinum) and all nonmetals (except oxygen and nitrogen) ignite in an atmosphere of

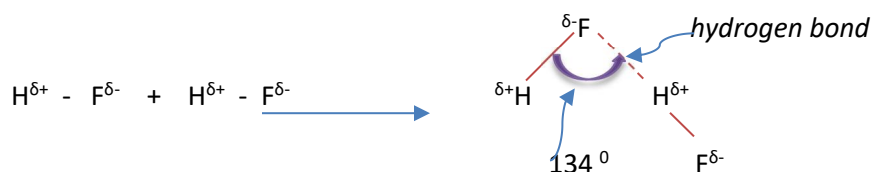
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fluorine at normal temperature. On heating, oxygen and nitrogen oxidize in fluorine, and platinum burns. Even materials like asbestos and bricks burn in fluorine. Fluorides of noble gases have been obtained: KrF_2 , KrF_4 , XeF_2 , XeF_4 , XeOF_2 , XeF_6 , XeOF_4 , and XeF_8 .

Fluorine displaces chlorine from chlorides: $\text{F}_2 + 2\text{NaCl} \rightarrow \text{Cl}_2 + 2\text{NaF}$

Hydrogen fluoride (**hydrofluoric**) acid (solution of HF in water) is a **weak acid**. Molecules of HF are bonded together by hydrogen bonds and form H_2F_2 . Hence, hydrofluoric acid can form acidic salts, for example KHF_2 .

When dissolved in water, hydrofluoric HF dissociates into H^+ and F^- . As a result hydrogen bonds are partly destroyed; hence, dissociation of HF into ions requires a substantial amount of energy.



Therefore, HF dissociates to a lesser degree (*since excess energy is needed to break strong hydrogen bond*) than other hydrogen halides.

The ion F^- , formed from dissociation of HF, bonds to a degree to undissociated HF, and HF_2^- ion is formed. Here, fluorine atoms are bonded to each other through hydrogen bond: $\text{F}^- + \text{HF} \rightleftharpoons \text{HF}_2^-$

Halide ions (negative ions of HHal) have reducing properties, therefore, when **HHal reacts on metals**, the latter (**metals**) can only be **oxidized** by hydrogen ions, H^+ .

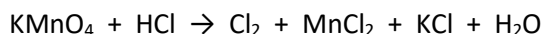
HF reacts with majority of metals. However, in many cases, **less soluble salts are formed**, in which on the metal surface, a **protective layer** is formed. That is how lead (Pb) behaves, and allows itself to be used for making apparatus which are resistant to the action of HF. Most salts of HF (fluorides) are less soluble in water; only fluorides of Na, K, Al, Sn and Ag dissolve well in water.

HF acid (concentrated) is poisonous and burns the skin. The **acid destroys glass**, reacts with silicon dioxide: $\text{SiO}_2 + 4\text{HF} \rightarrow \text{SiF}_4\uparrow + 2\text{H}_2\text{O}$

Since HF destroys glass, in the laboratory, **it is store in special plastic containers or in lead containers**.

Chlorine and its compounds

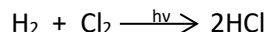
In nature, chlorine exists in form of chlorides. In industry, chlorine is produced together with hydrogen and sodium hydroxide during electrolysis of aqueous solution of sodium chloride. In the laboratory, chlorine is obtained from interaction of KMnO_4 with hydrochloric acid:



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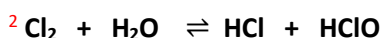
Chlorine – heavy gas, greenish-yellow colour with a pungent smell. Solution of 5 volume chlorine in 1 volume water is called **chlorine water**.

An active oxidizer, chlorine interacts with hydrogen in light with the formation of hydrogen chloride:



Hydrogen chloride – colourless gas with a pungent smell, dissolves well in water. In **1 volume** H_2O at 0°C , **507 volumes** of HCl dissolve. Solution of HCl in water is called **hydrochloric acid** – strong acid, its salts – chlorides.

When chlorine interacts with water or bases, **disproportionation** of chlorine takes place and besides HCl, oxygen – containing acids (oxy – acids) of chlorine are produced or their corresponding salts:



² Hydrolysis of Cl_2 does not go to the end. Therefore, chlorine water contains, besides Cl_2 , a significant amount of HCl and HOCl acids. If to chlorine water an alkali is added, then as a result of neutralization of HOCl and HCl acids, **equilibrium will shift to the right** in the system, $\text{Cl}_2 + \text{H}_2\text{O} \rightleftharpoons \text{HOCl} + \text{HCl}$. The reaction practically goes to the end, and a solution is obtained containing salts of HCl and HOCl:



The same result is obtained when Cl_2 is directly passed through alkali solution:



In ionic – molecular form: $\text{Cl}_2 + 2\text{OH}^- = \text{Cl}^- + \text{OCl}^- + \text{H}_2\text{O}$

The obtained solution is used for **bleaching**: $\text{KOCl} + \text{CO}_2 + \text{H}_2\text{O} = \text{KHCO}_3 + \text{HOCl}$

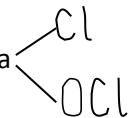


Salts of hypochlorous acid – hypochlorites – are obtained when **chlorine is passed through a cold solution of alkali**: $\text{Cl}_2 + 2\text{NaOH} = \text{NaCl} + \text{NaOCl} + \text{H}_2\text{O}$

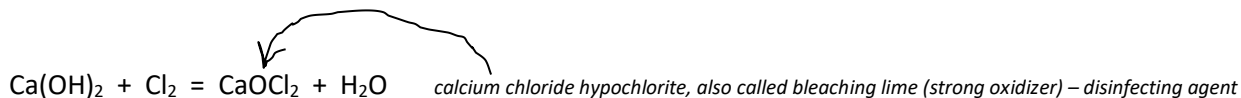
The obtained solution is called **Javel water**. It has oxidizing properties, it is used for whitening (bleaching) of cellulose (pulp, paper), clothe.

Whitening action of Javel water is connected to it absorbing CO_2 from air, which converts hypochlorite ions into molecules of weak hypochlorous acid: $\text{OCl}^- + \text{CO}_2 + \text{H}_2\text{O} \rightleftharpoons \text{HCO}_3^- + \text{HOCl}$

Molecules of HOCl are unstable and easily detaches oxygen (*atomic oxygen*), which whitens (bleaches) materials, oxidizing coloured ingredients: **$\text{HOCl} \rightarrow \text{HCl} + \text{O}$**

When Cl_2 acts on solid calcium hydroxide (slaked lime), a mixed salt is obtained Ca 

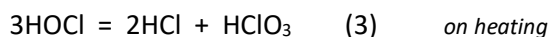
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Interaction of halogens with alkalis depends on temperature. E.g., when **chlorine is passed through a hot solution of KOH**, KClO_3 (berthollet's salt) is formed: $3\text{Cl}_2 + 6\text{KOH} \xrightarrow{\text{t}} \text{KClO}_3 + 5\text{KCl} + 3\text{H}_2\text{O}$

In solution, hypochlorous acid experiences three different types of change, which takes part independently:

$\text{HOCl} = \text{HCl} + \text{O}$ (1) *in cold solution*



By changing conditions, the reaction can be directed to any one direction.

By action of direct sunlight and presence of some catalysts or reducers, decomposition of hypochlorous acid takes place according to equation (1).

Reaction (2) takes place in the presence of water absorbing agents, for example, CaCl_2 . As a result, chlorine(I) oxide (hypochlorous anhydride) Cl_2O is obtained. It is an extremely unstable gas with a smell similar to that of chlorine.

Decomposition of HOCl according to reaction (3) easily takes place on heating. Therefore, if chlorine is passed through a hot solution of potassium hydroxide, then, in place of KOCl , KClO_3 is immediately obtained: $3\text{Cl}_2 + 6\text{KOH} = 5\text{KCl} + \text{KClO}_3 + 3\text{H}_2\text{O}$

Since **KClO_3 is less soluble in cold water**, then, when the solution is cooled, it falls as precipitate (**colourless crystals**).

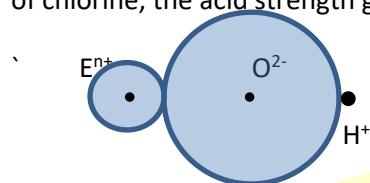
When KClO_3 is **heated**, it easily **decomposes**. In the presence of MnO_2 (as a catalyst), decomposition takes place according to equation: $2\text{KClO}_3 = 2\text{KCl} + 3\text{O}_2\uparrow$

With different **inflammable substances** (sulphur, carbon, phosphorus), KClO_3 forms a **mixture**, that **explodes** when hit. Therefore, it is used in artilleries.

Mixture of chlorine and calcium hypochlorite is called chloric or bleaching powder (white powder). Chlorine, in a quantity of 1.5g in 1m^3 of water is used for disinfection of water.

Bleaching acids

In the series of oxy – acids of chlorine: **HOCl , HClO_2 , HClO_3 , HClO_4** – with the increase in oxidation state of chlorine, the acid strength grows, but their oxidizing activity decreases.



oxidation state of $\text{Cl} \uparrow$ means O is held more strongly, therefore, decomposition $\downarrow$, as a result oxidizing activity $\downarrow$, acid is more stable, at the same time H^+ is easily detached, implying that acid strength $\uparrow$. The less stable the acid, the stronger it is an oxidizer.

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With more electronegative elements (fluorine, oxygen), chlorine displays the following oxidation states: +1(ClF , Cl_2O , HOCl), +3(ClF_3 , HClO_2), +5(ClF_5 , ClO_2F , HClO_3) and +7(ClOF_3 , ClO_2F_3 , ClO_3F , Cl_2O_7 , HClO_4).

KClO_3 , when heated in dry form (*colourless crystals*) detaches oxygen and vigorously oxidizes C in CO_2 , S in SO_2 , P in P_2O_5 . It is explosive; it is used in matchbox industry, in pyrotechnics (*craft of making fireworks*).

Chloric acid, HClO_3 , does not exist in a free state, since it disproportionate into ClO_2 and HClO_4 . In aqueous solution its amount cannot exceed 40%.

HClO_3 – strong oxidizer, strongly oxidizes organic substances. Its salts – chlorates – in acidic media also fully oxidizes organic substances, e.g.: $2\text{KClO}_3 + (\text{COOH})_2 + \text{H}_2\text{SO}_4 = \text{K}_2\text{SO}_4 + 2\text{CO}_2\uparrow + 2\text{ClO}_2\uparrow + 2\text{H}_2\text{O}$

Potassium chlorite (KClO_2) is a salt of chlorous acid, HClO_2 , which disproportionate even in dilute aqueous solution of and has a weak whitening property: $5\text{HClO}_2 = \text{Cl}_2 + 3\text{HClO}_3 + \text{H}_2\text{O}$

Chlorates, when carefully heated, disproportionate into compounds of chlorine with oxidation states, +7 and, -1: $4\text{KClO}_3 = 3\text{KClO}_4 + \text{KCl}$

Perchlorates (KClO_4) are the most stable oxy-compounds of chlorine. They are salts of perchloric acid, HClO_4 , the strongest of bleaching acids.

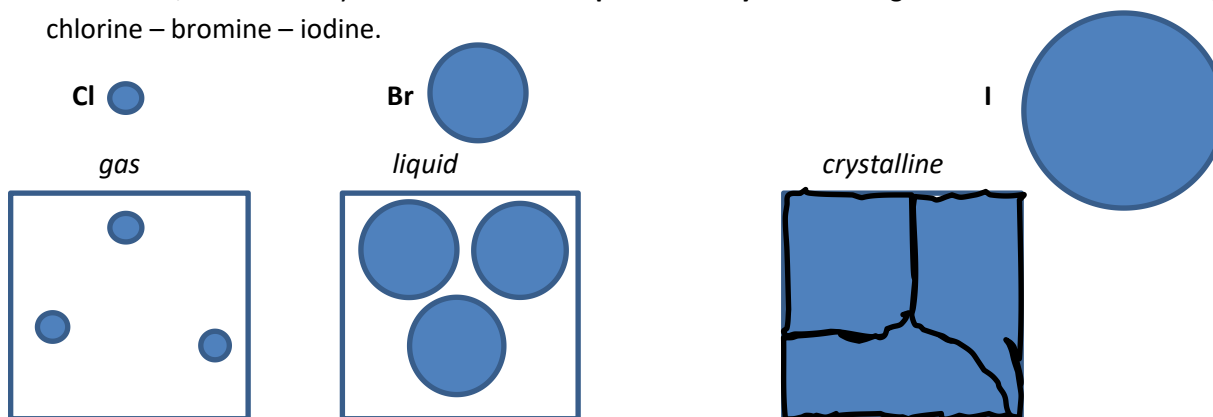
Organic compounds (e.g., sugar) mixed with KClO_3 , when reacted with conc. H_2SO_4 burns, since emission of HClO_4 and ClO_2 takes place: $3\text{KClO}_3 + 3\text{H}_2\text{SO}_4 = \text{HClO}_4 + 2\text{ClO}_2\uparrow + 2\text{H}_2\text{O} + 2\text{KHSO}_4$

Bromine, Iodine and their compounds

In nature, bromine and iodine are found in the form of salts NaBr , NaI , KBr , and KI . They are obtained by reacting chlorine with solutions of these salts: $2\text{NaBr} + \text{Cl}_2 \rightarrow 2\text{NaCl} + \text{Br}_2$, $2\text{NaI} + \text{Cl}_2 \rightarrow 2\text{NaCl} + \text{I}_2$

If chlorine is a **gas**, then, bromine is a dark brown **liquid** with a pungent smell, and iodine is a **crystalline** substance, dark coloured.

The change in the **phase – state** of halogens is caused by **increase** in the **intermolecular – dispersion** interaction, connected by increase in size and **polarizability** of the halogen molecules in the series; chlorine – bromine – iodine.



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Bromine and iodine interact with many metals: $2\text{Al} + 3\text{Br}_2 \rightarrow 2\text{AlBr}_3$, $2\text{Al} + 3\text{I}_2 \rightarrow 2\text{AlI}_3$

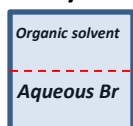
Oxidizing activity of halogens decreases from **Cl** to **Br** to **I**. So, interaction of **Br** with hydrogen takes place only at temperature $> 300^\circ\text{C}$, and interaction of **I** with hydrogen doesn't go to the end, it is a reversible process:

$$\text{H}_2 + \text{Br}_2 \xrightarrow{t > 300^\circ\text{C}} 2\text{HBr}, \quad \text{H}_2 + \text{I}_2 \rightleftharpoons 2\text{HI}$$

Cl easily oxidizes H, hence, easily interacts with it forming HCl. Br oxidizes H to a lesser extent, hence, will interact with difficulty.

Solutions of hydrogen bromide and hydrogen iodide are strong acids; their salts are very soluble in water. Only silver halides are less soluble in water: $\text{AgNO}_3 + \text{NaBr} \rightarrow \text{AgBr}\downarrow + \text{NaNO}_3$

Bromine and iodine are **very soluble in organic solvents** and can be **extracted** by these solvents from aqueous solutions.



shake well and Br will be extracted from aqueous solution

Solution of **Br** in CCl_4 is orange, and **I** – violet. With **starch**, iodine gives a **dark-blue** colouring (**characteristic reaction of iodine**), which disappears when heated and reappears when cooled.

Reducing properties of halide-ion increases in the series $\text{Cl}^- - \text{Br}^- - \text{I}^-$, and iodide-ion is a strong reducer.

Hal oxidizing ability $\downarrow$ down the group; Hal⁻ reducing ability $\uparrow$ down the group.

Bromine and iodine display positive oxidation states in their compounds with oxygen and more electronegative halogens.

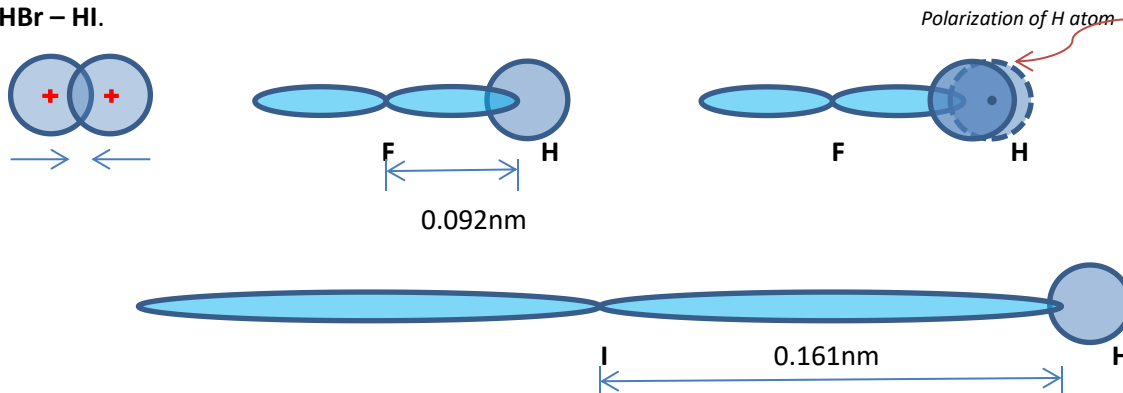
Hypobromous HOBr and hypoiodous HOI acids can only exist in strongly diluted solutions. They (acids) are weaker as electrolytes and oxidizers than is hypochlorous acid.

In the series $\text{HClO}_3 - \text{HBrO}_3 - \text{HIO}_3$, the acid strength falls.

General conclusion

The chemical bond in molecules of hydrogen halides is polaric covalent; common electron pair is shifted to the atom of halogen, the most electronegative.

The strength (stability) of the chemical bond in molecules of hydrogen halides fall in the series **HF – HCl – HBr – HI**.



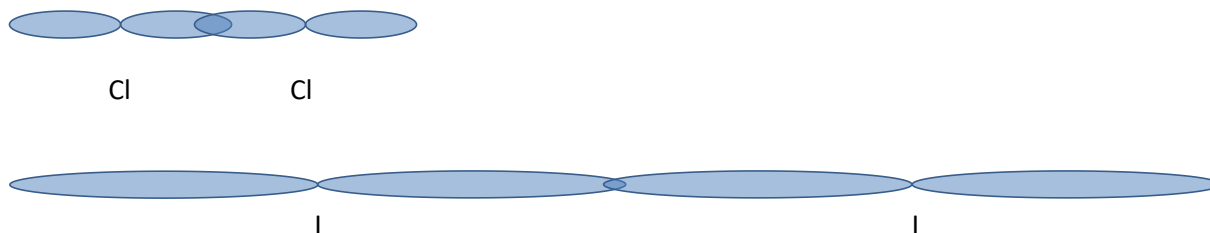
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- i) Electronegativity of halogen atom falls in the series $F - Cl - Br - I$. Hence, in HF, the electron cloud of hydrogen atom is displaced towards the halogen to a greater degree than in molecules HCl, HBr and HI, in which the displacement occurs less and less. From the diagrams above, in moving from HF to HI, the degree of overlapping of the electronic clouds of the atoms of hydrogen and halogen falls.
- ii) The region of overlapping is getting further and further away from the nucleus of the halogen atom.
- iii) There is strong shielding by electrons of intermediate levels.

All these reasons bring about the reduction in overlapping of interacting electronic clouds and as a result, weakening of the bond between atoms.

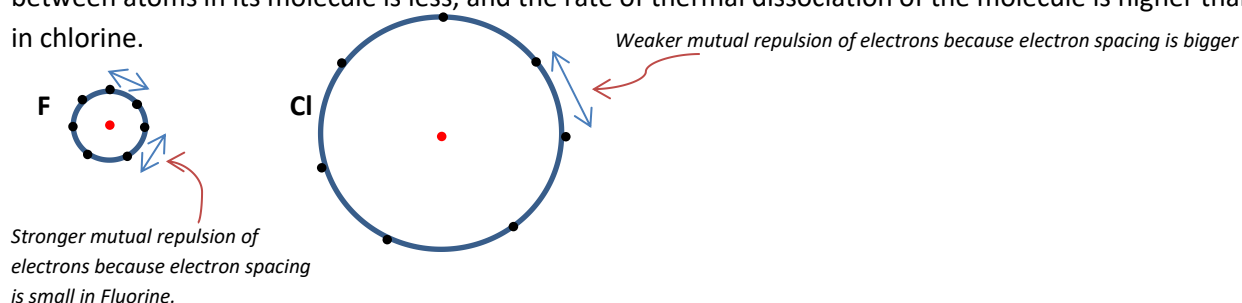
In the series **HI – HBr – HCl**, boiling and melting points change regularly (naturally), however in changing to HF, they sharply rise. This is because of association of HF molecules as a result of appearance of hydrogen bonds between them (HF). Near the boiling point, HF is in the form $(HF)_4$, only at temperature $90^\circ C$ does it become a simple molecule HF.

In the series **$Cl_2 - Br_2 - I_2$** , the bond strength between atoms in molecules slowly falls. Reason: With an increase in the size of the outer electronic clouds of interacting atoms, the degree of (their) overlapping reduces, and the region of overlapping is placed further and further away from the atomic nuclei.



Therefore, in moving from Cl to Br to I, attraction of the nuclei of the atoms of the halogens to the region of overlapping of the electronic clouds reduces. Besides, in the series $Cl - Br - I$, the number of intermediate electronic levels increases, in the process, shielding the nucleus, which in turn weakens the interaction of atomic nuclei with the region of overlapping.

At the same temperature, the rate of thermal dissociation of the molecules increases in moving from Cl_2 to Br_2 to I_2 . However, applying the same data, fluorine doesn't fall into this pattern: the bond strength between atoms in its molecule is less, and the rate of thermal dissociation of the molecule is higher than in chlorine.



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As a result of mutual repulsion of the valence electrons (many electrons) in fluorine atom, attraction of an extra electron to the nucleus (in order to form bond F – F) is weakened much more than in chlorine atom, which is bigger in size. Therefore, electron affinity in fluorine is less than in chlorine in forming of molecule Hal_2 . In chlorine atom, the bigger size (the number of valence electrons is same) means that there is less mutual repulsion of valence electrons, therefore, affinity of electron to the nucleus (in order to form bond Cl – Cl) is bigger, here, than in the case of fluorine.

Therefore, the bond energy in $\text{F}_2 < \text{Cl}_2$, but the bond energy in $\text{Cl}_2 > \text{Br}_2 > \text{I}_2$.

Note: In the case of HF, F is more electronegative so the electron pair is pulled towards F (the electron cloud of H is shifted towards F, as a result creating a bigger overlapping and a strong bond). In F_2 , the electron pair is exactly at the centre between the two nuclei. In order to form bond F – F, fluorine atom is reluctant to accept another electron due to repulsion of electrons on valence level.

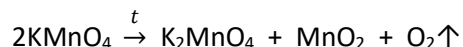
Group VIA elements

Group VIA forms four nonmetals: oxygen, sulphur, selenium, tellurium, which are called **Chalcogens**, and radioactive metal Polonium. Atoms of elements of group VIA have electronic formula ns^2np^4 . Their characteristic oxidation states are, -2, 0, +4, +6. In oxygen, there is an absence of d - orbitals, hence its valence is equal to two. The presence of d - orbitals in atoms of other elements allows them to have valencies of two, four or six.

Oxygen and its compounds

Oxygen is the most widespread element. It interacts with majority of chemical elements. Proteins, fats and hydrocarbons are organic oxygen – containing substances. Animals, plants all absorb oxygen from the atmosphere. With the participation of oxygen, mineralization of plant or animal organic remains takes place in the soil up to NH_3 , CO_2 and H_2O . Without aeration, the soil cannot be fertile.

Oxygen is a colourless gas, tasteless, odourless. Oxygen can be obtained by compression and separation of air. In the lab, oxygen can be produced by decomposition of KMnO_4 or potassium chlorate:



Two allotropic modifications of oxygen are known: oxygen O_2 and ozone O_3 .

Ozone can be produced from oxygen by action of electric charge or UV rays: $3\text{O}_2 \xrightarrow{h\nu} 2\text{O}_3$

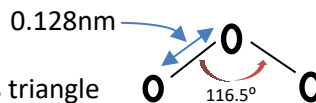
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Ozone – strong oxidizer, more than oxygen. Its activity is brought about by the breakup of its molecule with the formation of atomic oxygen, which is more active than molecular oxygen: $O_3 \rightarrow O_2 + O$

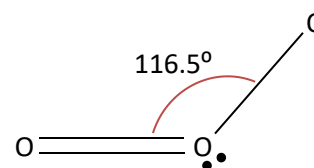
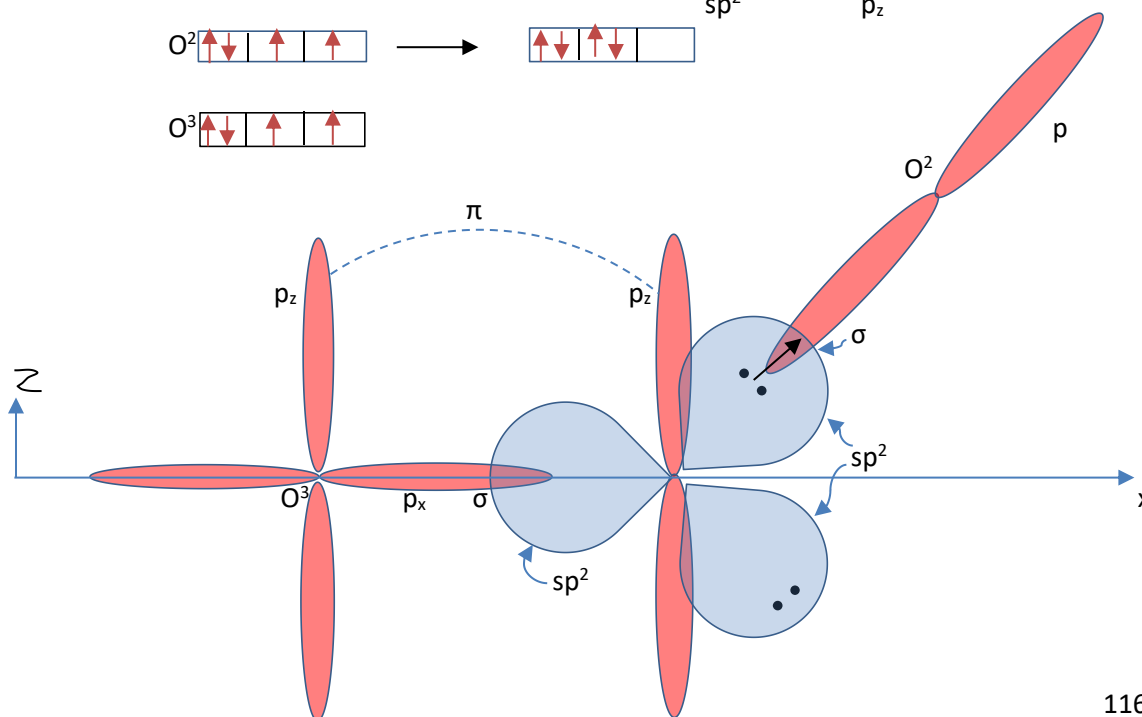
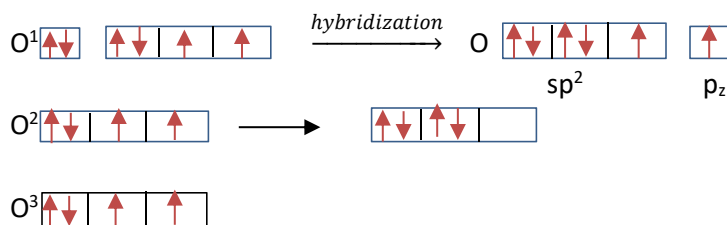
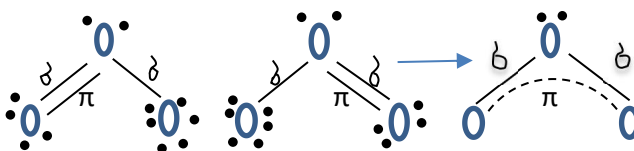
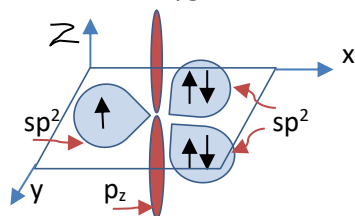
From solution of potassium iodide, ozone produces iodine: $2KI + H_2O + O_3 = I_2 + 2KOH + O_2$

If a paper soaked with a solution of KI and starch is placed in air that contains ozone, then the paper immediately turns blue. [*KI interacts with O_3 to give I_2 which in turn interacts with starch and turns blue*]. This reaction is used for identifying ozone.

A molecule of ozone is constructed in the form of isosceles triangle

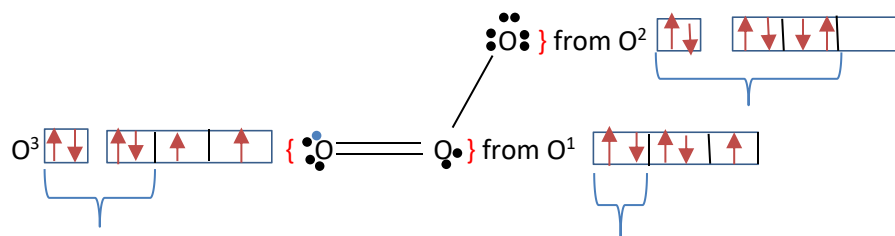


The central oxygen atom is in the state of sp^2 hybridization.



$116.5^\circ < 120^\circ$ because repels/squeezes other orbitals just like H_2O

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In the earth's atmosphere at a height of 25km exists the ozone layer, which absorbs strong shortwave radiations of the sun (λ shorter than 290nm), which protects all living things from destruction. In the air which we breathe, % volume of O_3 is 10^{-6} %. The small amount of ozone gives the air a pleasant fresh smell, but increase of ozone brings about discomfort and becomes life threatening. The allowed amount of ozone in the air is 0.8×10^{-5} % (volume).

Ozone is a bluish gas, water soluble, but more soluble in CCl_4 . A mixture containing more than 70% O_3 is explosive.

Active metals give O_3 molecule an electron without destroying the molecule, turning it into ozonide-ion, as a result of which ozonides are formed: $K + O_3 \rightarrow KO_3$

Ozone is used in preparing drinking water, for disinfecting sewage which contains cyanides, phenols; for bleaching some materials which are adequately stable to oxidation; for destroying smell in fats and oils and also as a deodorant in conditioning of air.

Sulphur and its compounds

In nature, Sulphur is found in a natural state and in the form of minerals, such as pyrite FeS_2 , gypsum $CaSO_4 \cdot 2H_2O$, mirabilite $NaSO_4 \cdot 10H_2O$. Sulphur is found in proteins of plants and animals. When meat, fish, eggs rot, hydrogen sulphide is evolved – with unpleasant smell. Compounds containing Sulphur are responsible for the bitter taste in onion and garlic. Elemental (simple) Sulphur is not toxic for human but has **fungicidal** properties.

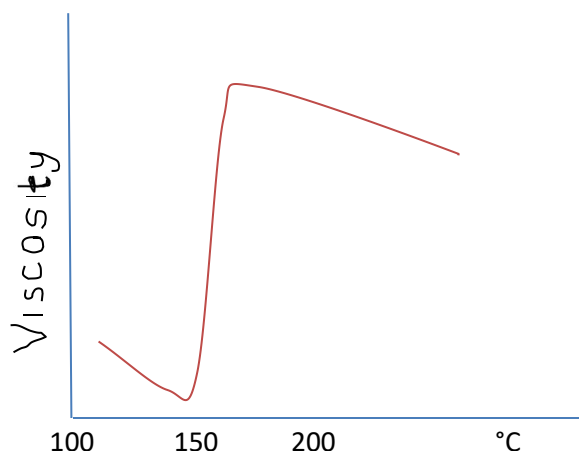
In industry Sulphur is obtained by purification, distillation of natural Sulphur or decomposition of pyrite: $FeS_2 \rightarrow FeS + S$.

Pure Sulphur is a fragile crystalline substance, yellow in colour. It has several allotropic modifications: rhombohedral, prismatic, plastic. Allotropy of Sulphur is caused by different structures of octatomic molecule S_8 . The molecule S_8 exists in molten Sulphur, molecule S_6 , S_4 , S_2 exist in vapour of Sulphur.

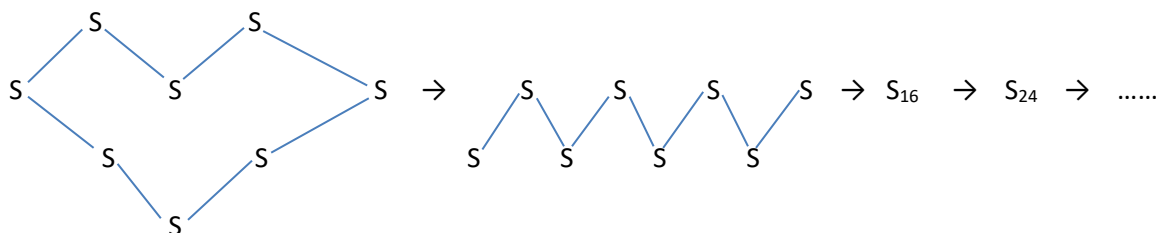
Heating of Sulphur results in very interesting changes in viscosity. At approximately $119^\circ C$, S melts to give a yellow liquid, whose viscosity gradually decreases until approximately $155^\circ C$. However, further

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heating causes the viscosity to increase very dramatically above 159 °C until the liquid pours very sluggishly. Above 200 °C, the viscosity again decreases, with the liquid eventually acquiring a more reddish hue (colour) at higher temperatures (diagram).



The explanation: tendency of S – S bonds to break and reform at high temperatures. Above 159 °C, the S₈ rings begin to open. The resulting S₈ chains can react with other S₈ rings to open them and form S₁₆ chains, and so on (diagram).

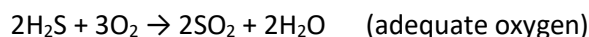


The longer the chain, the greater the viscosity (the more the chains can intertwine with each other). Large rings can also form by the linking of ends of chains. Chains exceeding 200 000 Sulphur atoms are formed at the temperature of maximum viscosity, near 180 °C. At higher temperatures, thermal breaking of S chains occurs more rapidly than propagation of chains, and average chain length decreases, accompanied by a decrease in viscosity.

Sulphur is a typical nonmetal. It reacts with metals, exhibiting oxidizing properties: $\text{Hg} + \text{S} \rightarrow \text{HgS}$. When reacting with more active nonmetals, Sulphur can exhibit reducing properties: $\text{S} + \text{O}_2 \rightarrow \text{SO}_2$. In lower oxidation states, -2, Sulphur forms a compound with hydrogen – H₂S, and with metals, e.g., Na₂S.

H₂S is a colourless poisonous gas with unpleasant smell. In aqueous solutions it behaves like a weak dibasic acid: $\text{H}_2\text{S} \rightleftharpoons \text{H}^+ + \text{HS}^-$ ($K_1 = 6 \times 10^{-8}$), $\text{HS}^- \rightleftharpoons \text{H}^+ + \text{S}^{2-}$ ($K_2 = 1 \times 10^{-15}$), K – dissociation constant.

H₂S has reducing properties. In air, it burns, oxidizing to Sulphur dioxide:



When there is inadequate oxygen, H₂S oxidizes to free Sulphur: $2\text{H}_2\text{S} + \text{O}_2 \rightarrow 2\text{S} + 2\text{H}_2\text{O}$ (inadequate).

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Such oxidizers as KMnO_4 , potassium dichromate ($\text{K}_2\text{Cr}_2\text{O}_7$), Cl , HNO_3 , oxidizes H_2S up to free S:



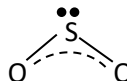
Sulphides of heavy metals are less soluble in water and have characteristic colours: ZnS – white, MnS – rose-coloured, CdS – yellow, Sb_2S_3 – orange, SnS – brownish, CuS – black.

In oxidation state, +4, S forms SO_2 , H_2SO_3 and corresponding salts, sulphites.

SO_2 is obtained industrially: $\text{S} + \text{O}_2 \rightarrow \text{SO}_2$, $\text{FeS}_2 + \text{O}_2 \rightarrow \text{SO}_2 + \text{Fe}_2\text{O}_3$

In the lab: $\text{Na}_2\text{SO}_3 + 2\text{HCl} \rightarrow 2\text{NaCl} + \text{H}_2\text{O} + \text{SO}_2 \uparrow$.

SO_2 molecule is constructed similar to ozone molecule:



When SO_2 is dissolved in water, dibasic H_2SO_3 acid forms: $\text{SO}_2 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_3$. Compounds of S in such intermediate oxidation state (H_2SO_3) can have oxidizing as well as reducing properties:



When SO_2 is oxidized by oxygen in the presence of a catalyst (Pt , V_2O_5 or NO), then SO_3 is formed.

Dissolving SO_3 in water leads to formation of H_2SO_4 : $\text{SO}_2 + \text{O}_2 \rightarrow \text{SO}_3$ $\text{SO}_3 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_4$

Sulphuric acid is a heavy, colourless, oily liquid. 100% H_2SO_4 is called monohydrate. Solution of SO_3 in monohydrate is called **Oleum** (fuming sulphuric acid) – $\text{H}_2\text{SO}_4 \cdot \text{SO}_3 = \text{H}_2\text{S}_2\text{O}_7$.

H_2SO_4 with water forms hydrates, absorbs water vapour, removing water from carbohydrates (sugar,...), charring (carbonizing) them: $\text{H}_2\text{SO}_4 (\text{conc.}) + \text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_4 \cdot \text{H}_2\text{O}$ H_2SO_4 energetically reacts with H_2O , releasing a lot of heat. Hence, always pour acid into water.

H_2SO_4 is an oxidizer. Dilute H_2SO_4 reacts with metals, standing in the potential series up to hydrogen and releases hydrogen gas: $\text{Zn} + \text{H}_2\text{SO}_4 (\text{dil.}) \rightarrow \text{ZnSO}_4 + \text{H}_2 \uparrow$.

Conc. H_2SO_4 reacts with majority of metals (except Au and Pt) with the formation of products of reduction of H_2SO_4 : SO_2 , free S or H_2S . The corresponding salt and water are also formed:

$\text{Cu} + \text{H}_2\text{SO}_4 (\text{conc.}) \rightarrow \text{CuSO}_4 + \text{SO}_2 + \text{H}_2\text{O}$ $\text{Zn} + \text{H}_2\text{SO}_4 (\text{conc.}) \rightarrow \text{ZnSO}_4 + \text{H}_2\text{S} + \text{H}_2\text{O}$ [*Zn is more active than Cu, hence stronger reducer*]

Conc. H_2SO_4 also oxidizes some nonmetals: $\text{H}_2\text{SO}_4 + \text{S} \rightarrow \text{SO}_2 + \text{H}_2\text{O}$ $\text{H}_2\text{SO}_4 + \text{C} \rightarrow \text{SO}_2 + \text{CO}_2 + \text{H}_2\text{O}$

Nonmetals are oxidized by conc. H_2SO_4 up to their oxides, and H_2SO_4 itself, is reduced to SO_2 .

Metal activity series: $\text{Li} > \text{K} > \text{Ba} > \text{Ca} > \text{Na} > \text{Mg} > \text{Be} > \text{Al} > \text{Zn} > \text{Cr} > \text{Fe} > \text{Sn} > \text{Pb} > \text{H} > \text{Cu} > \text{Hg} > \text{Ag} > \text{Pt} > \text{Au}$

Very active active less active passive

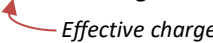
ACTION OF ACIDS ON METALS. Hydrochloric acid can oxidize substances only by H^+ ions, therefore, it (acid) oxidizes only active (Al, Zn) and moderately active (Fe, Sn) metals with the formation of their chlorides

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and release of free hydrogen. The low oxidizing ability of H^+ ions is displayed by the formation of metal ions with intermediate and not maximum charge: $CrCl_2$, $FeCl_2$, $SnCl_2$.

Very active metals (Na, K, Ba) are oxidized by water: $Ca + H_2O \rightarrow Ca(OH)_2 + H_2\uparrow$. Since O is more

electronegative than Cl, then, $\overset{+}{\text{H}}-\overset{-}{\text{O}}$, making "+" in H stronger. Hence, here, H is a slightly stronger oxidizer than in HCl.



Dilute (< 10 %) sulphuric acid acts on metals just like hydrochloric acid, i.e., in it (sulphuric), the role of oxidizer is played by H^+ ions, which is reduced to H_2 .

In concentrated acid, this role transfers to S atom (precisely to $\overset{+6}{\text{S}}\text{O}_4^-$ ions in the presence of H^+), therefore, the products of its reduction becomes $\overset{+4}{\text{S}}\text{O}_2\uparrow$, $\overset{0}{\text{S}}\downarrow$ and even $\overset{-2}{\text{H}_2\text{S}}\uparrow$, depending on the reducing activity of the metal.

Concentrated sulphuric acid energetically oxidizes metals, so much so that some of them (metals) are immediately covered by a thick oxide film protecting them from further oxidation. Passivation of the metal takes place. So, iron is passivated by sulphuric acid, if its concentration exceeds 75 %.

In a solution of nitric acid of any concentration, the starting oxidizer is always N atom (precisely $\overset{+5}{\text{N}}\text{O}_3^-$ ions in the presence of H^+), therefore, when it is reduced by metals, compounds of nitrogen are released: $\overset{+4}{\text{N}}\text{O}_2$ – from strong solutions, $\overset{+2}{\text{N}}\text{O}$ – from moderately diluted. In very dilute solutions, nitric acid is reduced by active metals up to NH_3 . In this case, ammonia remains in solution in a bonded state in the form NH_4NO_3 .

Nitric and concentrated sulphuric acids can oxidize metals up to higher oxidation states, than hydrochloric. For example, iron is oxidized up to $Fe(NO_3)_3$ and $Fe_2(SO_4)_3$.

Selenium, Tellurium and their compounds

Natural Se has six isotopes, of which the most widespread is ^{80}Se . Natural Te has eight isotopes of which the most widespread are ^{128}Te and ^{130}Te .

Tellurium is used in metallurgy. A little addition of Tellurium to lead increases lead's strength and resistance to corrosion. Copper and steel which contain Tellurium are easy to cut.

Selenium and Tellurium, like Sulphur, form compounds in which the oxidation state is, -2.

Hydrogen telluride $H_2\text{Te}$ and hydrogen selenide $H_2\text{Se}$ are gases with disgusting smell, very soluble in water and dissolves better than $H_2\text{S}$ does.

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On burning Se and Te, dioxides are formed – SeO_2 and TeO_2 . These are solid crystalline substances.

In the series $\text{SO}_2 - \text{SeO}_2 - \text{TeO}_2$, there is weakening of acidic properties, and TeO_2 can only be dissolved in a base: $\text{TeO}_2 + 2\text{KOH} = \text{K}_2\text{TeO}_3 + \text{H}_2\text{O}$. $\text{SO}_2/\text{SeO}_2 + \text{H}_2\text{O} \rightarrow \text{acid}$; $\text{TeO}_2 + \text{H}_2\text{O} \nrightarrow \text{acid}$.

The more unstable an oxide or oxy-acid is, the stronger it is an oxidizer³. [³Remember bleaching acids.]

SeO_2 – acidic oxide. TeO_2 – displays weak amphoteric properties.

SeO_2 – soluble in water, producing isomorphous (similar like acid to) H_2SO_3 , i.e., H_2SeO_3 .

TeO_2 – insoluble in water. To obtain acid of Te^{4+} : $3\text{Te} + \underbrace{4\text{HNO}_3}_{\text{moderate}} + \text{H}_2\text{O} = 3\text{H}_2\text{TeO}_3 + 4\text{NO}$ [look up for: nonmetal + HNO_3 (moderately dilute) in group VA elements].

On solubility in water, selenites are similar to sulphites, tellurites poorly dissolve in water.

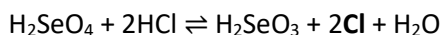
Selenious acid, H_2SeO_3 , can be evolved in the form of crystals, losing water at 70°C .

Tellurous acid, H_2TeO_3 , evolves as a precipitate of variable composition $\text{TeO}_2 \cdot n\text{H}_2\text{O}$. Both acids and their salts – selenites and tellurites – display stronger oxidizing properties than sulphurous acid and

sulphites, and can be reduced to simple substances: $\underbrace{\text{H}_2\text{SeO}_3}_{\text{strong oxidizer}} + 2\overset{+4}{\text{S}}\text{O}_2 + \text{H}_2\text{O} = \text{Se} \downarrow + 2\overset{+6}{\text{S}}\text{O}_4$

A similar process occurs with H_2TeO_3 . These reactions are the basis for industrial production of Se and Te.

Selenium trioxide, SeO_3 , energetically interacts with water forming selenic acid H_2SeO_4 . This is a solid substance, in solution it reminds us of sulphuric, but it is a stronger oxidizer:



A mixture of selenic and hydrochloric acids dissolve gold and platinum due to the formation of **atomic chlorine**: $6\text{H}_2\text{SeO}_4 + 12\text{HCl} + 4\text{Au} = 6\text{H}_2\text{SeO}_3 + 4\text{AuCl}_3 + 6\text{H}_2\text{O}$

Telluric acid H_6TeO_6 , $\underbrace{[\text{Te}(\text{OH})_6]}_{\text{hydroxide of nonmetal is acid}}$, hexabasic acid, trihydrate $\text{TeO}_3 \cdot 3\text{H}_2\text{O}$ – is a crystalline substance, very soluble in hot water. It is a very weak acid and also a weak oxidizer. It is a less active oxidizer than H_2SeO_4 .

All compounds of selenium are poisonous. The minimum allowed dosage in the air at place of work is $2 \times 10^{-4} \text{ g/m}^3$.

The limited amount of dosage allowed in the air for tellurium is 10^{-5} g/m^3 . The most dangerous of the compounds of tellurium is H_2Te .

When H_2SeO_3 and H_2TeO_3 react with strong oxidizers, H_2SeO_4 and H_2TeO_4 [$\text{H}_6\text{TeO}_6 = \text{H}_2\text{TeO}_4 + 2\text{H}_2\text{O}$] acids are produced: $\text{H}_2\text{SeO}_3 + \text{H}_2\text{O}_2 = \text{H}_2\text{SeO}_4 + \text{H}_2\text{O}$ $3\text{H}_2\text{TeO}_3 + 2\text{HClO}_4 = 3\text{H}_2\text{TeO}_4 + \text{Cl}_2 + \text{H}_2\text{O} + 2\text{O}_2$

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Selenic acid is similar to sulphuric, their salts are isomorphous. Selenic is less stable than sulphuric and has a stronger oxidizing action than sulphuric.

Telluric acid in structure and properties differs from selenic. In crystalline state it has composition $\text{Te}(\text{OH})_6 - \text{H}_6\text{TeO}_6$.

 A compound which contains in it hydroxo group is called hydroxide. Hydroxide of a metal is a base, hydroxide of a nonmetal is an acid.

Telluric is a very weak acid and also weak oxidizer.

SeO_3 and TeO_3 are crystalline substances. SeO_3 is soluble in water, less stable, and, therefore, a strong oxidizer. TeO_3 is insoluble in water, unlike SeO_3 , it is quite a stable substance.

Oxygen and Sulphur are the main ore-forming elements. Their most important derivatives are water, sulphuric acid and sulphide. Selenium, selenides and tellurides are semiconductors (used in electronic industry).

Group VA elements

Group VA elements have electronic formula ns^2np^3 .

Nitrogen, Phosphorous and Arsenic are nonmetals, Bismuth and Antimony have metallic properties.

The most characteristic oxidation states are +5, +3, -3.

Oxides E_2O_5 have acidic properties, oxides E_2O_3 have acidic properties – in the case of N and P, amphoteric – in the case of As and Sb, and basic – in the case of Bi.

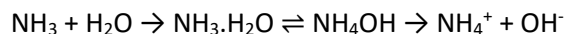
Nitrogen and its compounds

Nitrogen is found in nature basically in a free state in the atmosphere – 78% N_2 . Bonded nitrogen accumulates in plant and animal world – in organs of animals. All important parts of the cells are made from protein. Without protein, no life, and without nitrogen, no proteins. Industrially, N is obtained by compression of air followed by heating, in which N first evaporates (b.p. = -195.8°C) and then later oxygen (b.p. = -183.0°C).

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Nitrogen is used in industry basically for producing nitrogen fertilizer, in part for the synthesis of ammonia: $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$.

A compound of nitrogen with oxidation state, -3, ammonia is a colourless gas with a pungent smell. Ammonia is poisonous, lighter than air, very soluble in water⁴ with the formation of ammonium hydroxide, which is taken to be a weak base ($K = 1.79 \times 10^{-5}$):



⁴Solubility of NH_3 in water is better than any other gas: 700 vol. in 1 vol. H_2O . This is because of acceptance of a proton by N which has unshared pair of electrons.

The basic properties of ammonia are displayed in its reactions with acids: $\text{NH}_3 + \text{HCl} \rightarrow \text{NH}_4\text{Cl}$ (salt)

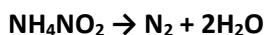
Ammonia has reducing properties. (N has the lowest oxidation state)

Ammonia burns in air, with oxygen, to give nitrogen or in the presence of platinum catalyst, to give NO:



The presence of unshared pair of electrons in the N of NH_3 is the reason for the donor property of NH_3 , to form complex compounds – ammoniates: $\text{CuSO}_4 + 4\text{NH}_3 \rightarrow [\text{Cu}(\text{NH}_3)_4]\text{SO}_4$

Salts of ammonia decompose on heating: $\text{NH}_4\text{Cl} \rightarrow \text{NH}_3 + \text{HCl}$



In the last reaction, we obtain nitric(I) oxide – N_2O , called “laughing gas”. N_2O is a gas with sweet taste, pleasant smell. It has intoxicating action on a person. It is used as an anesthesia in dentistry. It is a neutral oxide⁵, decomposing to N and O on heating.

⁵There are two types of oxides – salt forming and non-salt forming. Salt forming – basic, acidic, amphoteric.

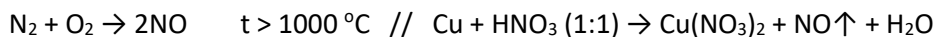
Basic – oxides that form salts when they react with acids or acidic oxides, e.g., CaO , $\text{Ca}(\text{OH})_2$.

Acidic – oxides that form salts when they react with bases or basic oxides. By attaching water directly or indirectly, acidic oxides form acids. E.g., SiO_2 forms silicic acid H_2SiO_3 , N_2O_5 forms nitric acid HNO_3 . Acidic oxides can be obtained by removing water from the relevant acids, which is why they are also known as acid anhydrides.

Amphoteric – oxides that form salts when they react either with acids or bases, e.g., ZnO , Al_2O_3 .

Non – salt-forming oxides react neither with acids nor bases, e.g., N_2O and NO .

Nitric(II) oxide, which is also a neutral oxide, is obtained according to reactions:



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Nitric(II) oxide is a colourless gas, a reducer and is easily oxidized by oxygen of the air up to nitric(IV) oxide: $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$

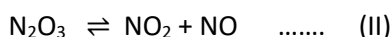
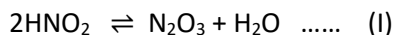
Nitric(IV) oxide, NO_2 , is a brown gas, poisonous. On cooling, it dimerizes into colourless oxide N_2O_4 .

NO_2 is a strong oxidizer: carbon, phosphorous and Sulphur burn in it.

NO_2 interacts with water and bases, displaying acidic properties, while N^{+4} disproportionate to N^{+5} and N^{+3} : $2\text{NO}_2 + \text{H}_2\text{O} \rightarrow \text{HNO}_3 + \text{HNO}_2$ $2\text{NO}_2 + 2\text{NaOH} \rightarrow \text{NaNO}_3 + \text{NaNO}_2 + \text{H}_2\text{O}$

Nitric(III) oxide, **N_2O_3** , exists only at low temperatures. At $t = 3.5^\circ\text{C}$, it boils and decomposes: $\text{N}_2\text{O}_3 \rightarrow \text{NO}_2 + \text{NO}$. This oxide corresponds to weak nitrous acid **HNO_2** , which exists only in dilute solutions and easily decomposes: **$3\text{HNO}_2 \rightarrow \text{H}_2\text{O} + \text{HNO}_3 + 2\text{NO}$**

Instability of HNO_2 rises with an increase in concentration of the solution. When a strong acid acts on a salt of nitrous acid (nitrite), the formed HNO_2 releases N_2O_3 (the solution turns blue), which then decomposes to NO_2 and NO :



NO_2 with water forms two acids: $2\text{NO}_2 + \text{H}_2\text{O} \rightleftharpoons \text{HNO}_2 + \text{HNO}_3 \quad \dots\dots \quad (\text{III})$

Summing up equations I, II, III, we obtain a general equation of the process of decomposition:



Combining (I) and (II): $2\text{HNO}_2 = \text{NO}_2 + \text{NO} + \text{H}_2\text{O} \quad \dots\dots \quad (\text{IV})$

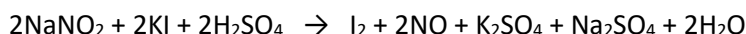
From (III): $2\text{NO}_2 = \text{HNO}_2 + \text{HNO}_3 - \text{H}_2\text{O} \quad \dots\dots \quad (\text{V})$

Multiplying equation (IV) by 2: $4\text{HNO}_2 = 2\text{NO}_2 + 2\text{NO} + 2\text{H}_2\text{O}$

Putting (V) in the above equation: $4\text{HNO}_2 = \text{HNO}_2 + \text{HNO}_3 - \text{H}_2\text{O} + 2\text{NO} + 2\text{H}_2\text{O}$



Salts of nitrous acid (nitrites) are stable, they have oxidizing as well as reducing properties (N is in intermediate state):



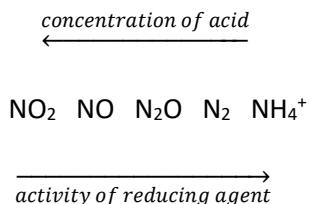
The hydrate compound of nitric(V) oxide N_2O_5 is nitric acid HNO_3 . Obtaining HNO_3 and its salts, nitrates, in industry is based on i) the synthesis of ammonia, ii) its oxidation to NO , iii) oxidation of NO to NO_2 and iv) interaction of NO_2 with water in the presence of air oxygen:

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- i) $3\text{H}_2 + \text{N}_2 \rightarrow 2\text{NH}_3$
- ii) $4\text{NH}_3 + 5\text{O}_2 \rightarrow 4\text{NO} + 6\text{H}_2\text{O}$
- iii) $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$
- iv) $4\text{NO}_2 + \text{O}_2 + 2\text{H}_2\text{O} \rightarrow 4\text{HNO}_3$

Nitric acid is a strong acid, its salts are very soluble in water. HNO_3 and its salts, nitrates, are oxidizers (N is in the highest oxidation state).

Composition of products of HNO_3 reduction depends on the activity of the reducing agent and concentration of the acid; reduction of N proceeds further when the reducing agent is more active and dilution of acid is greater:



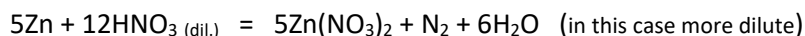
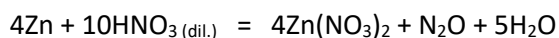
Consequently, when conc. HNO_3 reacts with nonmetals or with metals having a low activity, nitrogen dioxide, NO_2 , is formed: $\text{P} + 5\text{HNO}_3 = \text{H}_3\text{PO}_4 + 5\text{NO}_2 + \text{H}_2\text{O}$ nonmetals transform into acids



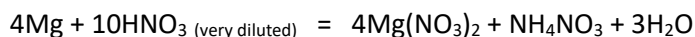
Conc. HNO_3 oxidizes most elements to their maximum oxidation state.

When more dilute (moderately concentrated) nitric acid reacts with metals having a low activity, nitrogen monoxide maybe liberated: $3\text{Cu} + 8\text{HNO}_3 (35\%) = 3\text{Cu}(\text{NO}_3)_2 + 2\text{NO} + 4\text{H}_2\text{O}$ (also with P)

When active metals are involved, the products include dinitrogen oxide or free nitrogen:



Much more diluted nitric acid when reacted with active metals may be reduced to ammonium ion NH_4^+ , which forms ammonium nitrate with the acid:



Conc. HNO_3 + nonmetal $\rightarrow$ acid + NO_2 + H_2O // moderately dil. HNO_3 + nonmetal + $\text{H}_2\text{O} \rightarrow$ acid + NO

Conc. HNO_3 does not react with Au and Pt, passivates Fe, Cr and Al.

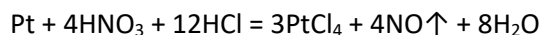
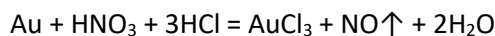
Very passive metals like Au and Pt can be dissolved by a mixture of one volume of conc. HNO_3 with three volumes of strong HCl. This mixture is called "**aqua regia**", the oxidizer in it is atomic chlorine, which is formed as a result of the reaction:

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$\text{HNO}_3 + 3\text{HCl} \rightleftharpoons 2\text{H}_2\text{O} + \text{NOCl} + \text{Cl}_2$, and is released from nitroze chloride ($\text{NOCl} = \text{NO} + \text{Cl}$). [Remember: $\text{H}_2\text{SeO}_4 + 2\text{HCl} \rightleftharpoons 2\text{Cl} + \text{H}_2\text{SeO}_3 + \text{H}_2\text{O}$, also used for dissolving Au and Pt].

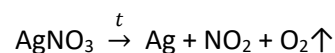
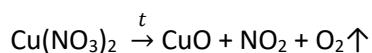
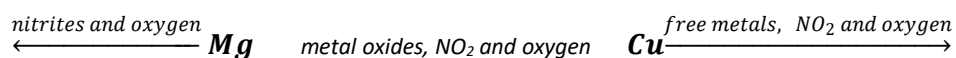
Oxidation of these metals (Au, Pt) is also assisted by complex formation of the products of their oxidation with chloride – ions.

Reaction of oxidation of gold and platinum:



With excess HCl, gold(III) chloride and platinum(IV) chloride form complex compounds $\text{H}[\text{AuCl}_4]$ and $\text{H}_2[\text{PtCl}_6]$.

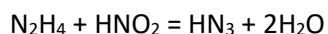
Nitrates decompose on heating. The contents of the products depend on the position of the metal of the salt in the activity series. When metal nitrates standing to the left of Mg in this series decompose, nitrites and oxygen are produced, from Mg to Cu – metal oxides, nitric(IV) oxide and oxygen, standing after Cu – free metals, nitric(IV) oxide and oxygen.



Salts of HNO_3 and salts of NH_3 are important mineral fertilizers.

Careful oxidation of NH_3 by hypochlorite NaOCl produces hydrazine N_2H_4 ($\text{H}_2\text{N} - \text{NH}_2$), where oxidation state of N is, -2. [$2\text{NH}_3 + \text{NaOCl} = \text{N}_2\text{H}_4 + \text{NaCl} + \text{H}_2\text{O}$]. N_2H_4 is a colourless liquid, it mixes in any proportion with water. Being a reducer, it burns in air (oxidizing to N_2), releasing a lot of heat. With the help of an energetic reducer (atomic hydrogen), it can be reduced to NH_3 . N_2H_4 and its derivatives are poisonous. They are used in organic synthesis in producing insecticides, plastics and explosives.

Hydrazine can be oxidized by nitrous acid (inadequate) up to hydrazoic acid, HN_3 ($\text{H} - \text{N} - \text{N} \equiv \text{N}$):



In aqueous solution, HN_3 dissociates to ions H^+ and N_3^- . Hydrazoic anion, N_3^- , has a linear structure $[\text{:}\ddot{\text{N}} = \text{N} = \ddot{\text{N}}\text{:}]^-$. By what type of hybridization of the central N's atomic orbitals is the formation of the hydrazoic anion? Sketch the anion using clearly labeled orbitals.

The salts of hydrazoic acid – azides – are thermodynamically more unstable, the more passive the metals contained in them are. And azides like AgN_3 , $\text{Pb}(\text{N}_3)_2$ decompose with an explosion when hit and are used as detonators.

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Phosphorous and its compounds

Phosphorous – solid constituent of plant and animal proteins. In plants it is concentrated in seeds, in animals – nerve tissue, muscles and skeleton.

Human organs contain about 1.5 kg of phosphorous: 1.4 kg in the bones, 130 g in the muscles and 12 g in the nerve tissue.

In nature, P is found in bonded form. The most important minerals of P are apatite $\text{Ca}_5(\text{PO}_4)_3\text{F}$ and phosphorite $\text{Ca}_3(\text{PO}_4)_2$. P can be obtained by heating a mixture of phosphorite, carbon and sand in a special furnace:

$$\text{Ca}_3(\text{PO}_4)_2 + 5\text{C} + 3\text{SiO}_2 \xrightarrow{t} 2\text{P} + 3\text{CaSiO}_3 + 5\text{CO}$$

P produces several allotropic modifications⁶: white, black and red phosphorous.

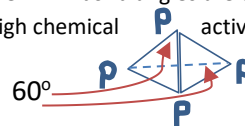
⁶Let's take a piece of phosphorous – it is a white translucent substance (waxy solid), with m.p. = 44.2 °C and very poisonous; in air and in the dark, P glows and can self-ignite.

P is a simple substance, it cannot be separated into other substances. However, if P is heated without availability of air, then after some time its properties sharply change: P attains a red-violet colour (brittle powder), it stops glowing in the dark, and it is made non-poisonous and does not ignite in air, besides, these new properties do not disappear when heating is stopped. Hence, undoubtedly, a change from one substance to another takes place, but the change is unique; the substance taken originally does not decompose, and nothing is bonded to it. Therefore, both substances – the original and the one obtained after heating – are simply different forms of existence of one and the same element – phosphorous, in the free state, the first one is called white, the second – red phosphorous.

The proof that white and red phosphorous are truly different forms of one and the same element and contain the same atoms, is their relation to oxygen: when heated in oxygen, both white and red phosphorous interact with it, forming one and the same substance – phosphoric anhydride.

White P is obtained in solid form when phosphorous vapour is suddenly cooled. In the cold, white P is brittle, but at temperatures greater than 15 °C, it becomes soft and is easily cut with a knife.

White phosphorous – P_4 – the stability of the bonds between these atoms is low since the P-P-P bond angles are 60° in each P_4 unit. There is considerable amount of strain in the bonds which makes itself felt in the high chemical activity of P.



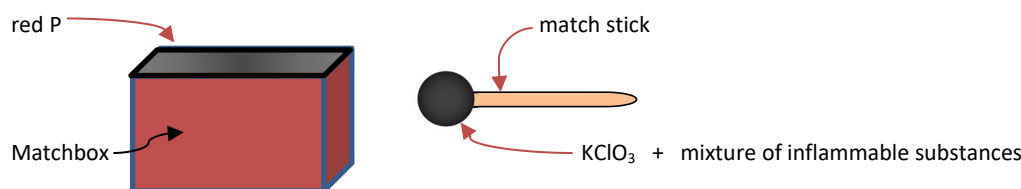
Red phosphorous – when strongly heated, sublimates. On cooling the vapours, white phosphorous is obtained. White P very slowly changes into red in the course of many years. This change is accelerated by raising the temperature – heating to 270 °C for several days (in vacuum).

Black phosphorous – is obtained from white when it is heated up to 200 – 220 °C under very high pressure in the absence of air. In form it is similar to graphite.

Phosphorous is used in the production of organophosphorus pesticides (Malathion insecticide, Thiophos, Trichlorfon).

Red phosphorous is used in the manufacture of matchboxes. It is contained on the surface of the matchbox (diagram). The head of the match stick has a mixture of inflammable substances with KClO_3 (Berthollett's salt).

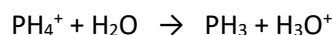
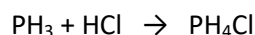
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Phosphorous in oxidation state, -3, forms hydrogen compound phosphine, PH_3 , which is analogic to ammonia. PH_3 is only slightly soluble in water. This is because P atom is not sufficiently electronegative to hydrogen bond to the hydrogen of a H_2O molecule. The presence of hydrogen bonding in NH_3 and its absence in PH_3 also explains why PH_3 , which is a heavier molecule than NH_3 , has a lower boiling point.

The oxidation state, -3, is less characteristic of P than it is for N. **The stable oxidation state for P is, +5.**

Basic properties of PH_3 are weaker than those of NH_3 because PH_4^+ ion is decomposed by H_2O . Salts of phosphonium ion in aqueous solutions are unstable:



PH_3 forms salts only with the strongest acids.

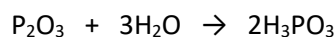
Phosphine has reducing properties and is a stronger reducer than ammonia.

Phosphine burns in air: $2\text{PH}_3 + 4\text{O}_2 \rightarrow \text{P}_2\text{O}_5 + 3\text{H}_2\text{O}$

Phosphine is a poisonous gas, with garlic smell. It can be obtained from zinc phosphide by reacting with acids or water: $\text{Zn}_3\text{P}_2 + 6\text{HCl} \rightarrow 2\text{PH}_3 + 3\text{ZnCl}_2$

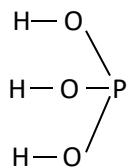
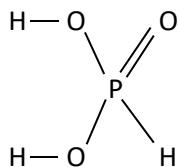
Zinc phosphide is used as a rodent poison.

Phosphorus(III) oxide P_2O_3 (or P_4O_6) forms during the oxidation of P in inadequate oxygen, and it is a white crystalline substance. This acidic oxide interacts with water, forming phosphorous acid:



Phosphorus acid in aqueous solutions exist in two tautomeric forms (interchangeable forms):

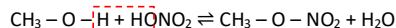
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Common properties of spirits and phenols is the mobility

of hydrogen of the hydroxyl group OH. Spirit & phenols

interact with acids forming

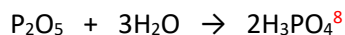


Spirit + Acid $\rightarrow$ Ester

Under the action of bases, equilibrium shifts to the left, upon the action of spirits ROH – to the right. Therefore, phosphorous acid is a dibasic weak acid which forms acidic salt NaH_2PO_3 if acted upon by bases, e.g., NaOH, and also form intermediate salt Na_2HPO_3 . Triple displaced phosphates are not known.

Phosphorous(V) oxide P_2O_5 (or P_4O_{10}) is obtained when P is heated in air: $4\text{P} + 5\text{O}_2 \rightarrow 2\text{P}_2\text{O}_5$

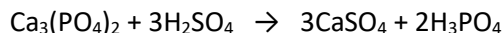
Solid crystalline substance, P_2O_5 , is hygroscopic and is used as a dehydrating agent. On reacting with water, P_2O_5 forms in succession metaphosphoric acid HPO_3 and orthophosphoric acid H_3PO_4 .⁷



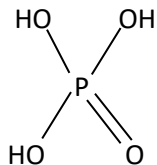
⁷When an element in the same oxidation state forms several acids each containing one atom of the given element in a molecule (e.g., HPO_3 and H_3PO_4), the prefix *meta*- is used with the name of the acid containing the smallest number of oxygen atoms, and the prefix *ortho*-, with the name of the one containing the greatest number of oxygen atoms.

⁸ H_3PO_4 – not oxidizer. H_3PO_3 – strong reducer. Oxidation state, -3, is less characteristic for P.

In industry, phosphoric acid is produced when H_2SO_4 reacts with phosphorites:



Orthophosphoric acid is a crystalline substance (m.p. = 42 °C), soluble in water. Like a tribasic acid of medium strength, it dissociates in steps.



A distinction is made between medium salts – phosphates (Na_3PO_4) [triple displaced] and acid salts – hydrophosphates (Na_2HPO_4) [double displaced] and dihydrophosphates (NaH_2PO_4) [single displaced]. Only phosphates and hydrophosphates of alkali metals and ammonia are soluble in water. All dihydrophosphates are soluble in water.

Many phosphates are used as fertilizers (mineral): phosphorite fertilizer $\text{Ca}_3(\text{PO}_4)_2$, CaHPO_4 (on acidic soils), superphosphate $\text{Ca}(\text{H}_2\text{PO}_4)_2 \cdot 2\text{CaSO}_4$, binary superphosphate $\text{Ca}(\text{H}_2\text{PO}_4)_2$.

Mixture of CaSO_4 and calcium dihydrophosphate $\text{Ca}(\text{H}_2\text{PO}_4)_2$ obtained when $\text{Ca}_3(\text{PO}_4)_2$ interacts with H_2SO_4 .

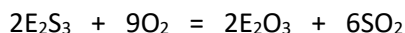
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Arsenic, Antimony, Bismuth and their compounds

As, Sb and Bi are found in the earth's crust in oxidized state, mainly in the form of sulphides.

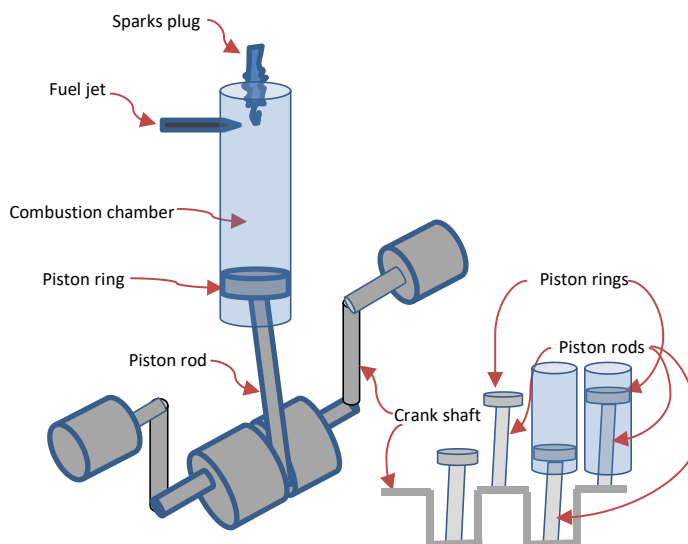
Arsenic is found in all plant and animal organs. In human body it is mostly in the liver (from 2 to 12 mg in 1 kg); it is assumed that micro doses of As increase the resistance of organs against deadly bacteria.

These elements in the form of simple substances are obtained by heating their natural sulphides:



Grey As is used as an additive in some alloys of coloured metals. Up to 0.45% As increases the breaking strength of copper, gives it hardness and increases corrosive resistance; up to 1% As increases hardness of lead.

About 200 alloys contain Sb; it gives hardness to lead and tin. Alloys of Sb (up to 15%) with tin and addition of lead, sometimes Cu, Zn, Bi, have antifriction properties and are poured in piston bearings (diagram).

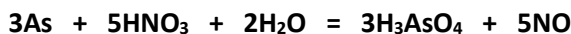


Bismuth is the basis of many fusible (low melting) alloys. A bigger % of Bi is contained in their alloys.

In the compounds of these elements, oxidation states of, -3, +3 and +5 are displayed, of which the most stable for Bi is, +3 (showing metallic characteristic).

As, Sb and Bi are not soluble in HCl, and in relation to HNO_3 , they display increasing metallic characteristics from As to Bi.

If Arsenic behaves like a nonmetal, forming higher (oxidation state, +5) oxygen containing (arsenic) acid H_3AsO_4 :



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And Antimony, also in moving to oxidation state, +5, gives non-soluble hydroxide⁹, called β – antimonite acid HSbO_3 (correctly $\text{Sb}_2\text{O}_5 \cdot n\text{H}_2\text{O}$): $\boxed{3\text{Sb} + 5\text{HNO}_3 = 3\text{HSbO}_3\downarrow + 5\text{NO} + \text{H}_2\text{O}}$

⁹ $\text{Sb}(\text{OH})_5 = \text{HSbO}_3 + 2\text{H}_2\text{O}$ Sb – less as nonmetal coming from: $\text{Sb} + \text{HNO}_3 + \text{H}_2\text{O} = \text{HSbO}_3\downarrow + 2\text{H}_2\text{O} + \text{NO}$

Then Bismuth, after oxidizing to Bi^{3+} , changes into Bismuth(III) nitrate:



As and Sb reacts with conc. HNO_3 , but Bi is passivated by it. Bi dissolves in dilute solution of HNO_3 .

With the majority of metals, As, Sb and Bi form intermetallic compounds, but only with very active metals of groups I and II do they give salt-forming substances (type Ca_3E_2), called arsenides, stibides (antimonides) and bismutides, which disintegrate in acidic media with the formation of arsine, stibine and bismutine, respectively (EH_3): $\text{Ca}_3\text{E}_2 + 6\text{HCl} = 3\text{CaCl}_2 + 2\text{EH}_3\uparrow$

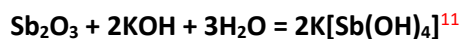
In relation to the strengthening of metallic character from As to Bi, the stability of EH_3 in the series $\text{AsH}_3 - \text{SbH}_3 - \text{BiH}_3$ reduces. Note: “negative” oxidation state is not characteristic of metals (Bi).

As easily oxidizes in air, Sb needs more, and Bi – strong heating to change into E_2O_3 . Metals have a thin oxide film that must be broken to enable interaction with oxygen, hence, as we move from nonmetal to metal there is difficulty for these elements to oxidize.

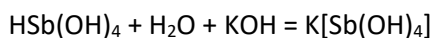
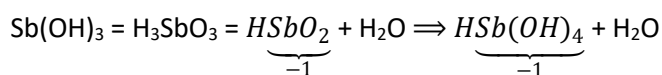
Arsenic(III) oxide As_2O_3 – white arsenic – is soluble in water and alkalis, i.e., has acidic character. Consequently, the hydrate H_3AsO_3 exists only in solution and is called arsenous acid¹⁰.

¹⁰ $\text{As}_2\text{O}_3 + 3\text{H}_2\text{O} = 2\text{As}(\text{OH})_3$ (arsenic hydroxide) $\equiv 2\text{H}_3\text{AsO}_3$ – it is amphoteric but leaning more towards acidic: $\text{As}_2\text{O}_3 + 6\text{KOH} = 2\text{K}_3\text{AsO}_3 + \text{H}_2\text{O}$ [$\text{As}(\text{OH})_3 - \text{H}_3\text{AsO}_3 + \text{KOH} = \text{K}_3\text{AsO}_3$]

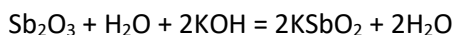
Antimony(III) oxide Sb_2O_3 – is practically not soluble in water, it reacts with solutions of acids and alkalis: $\text{Sb}_2\text{O}_3 + 6\text{HCl} = 3\text{SbCl}_3 + 3\text{H}_2\text{O}$



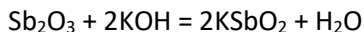
¹¹Since the product is hydroxocomplex, start with hydroxo acid $\text{HSb}(\text{OH})_4$:



If the product was not hydroxocomplex but oxo-complex, then we would have:



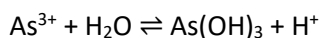
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Bismuth(III) oxide Bi_2O_3 – dissolves in strong acids with the formation of salts of type BiCl_3 and with difficulty dissolves in concentrated solutions of alkalis. Hence, it is thought that Bi_2O_3 and Bi(OH)_3 possess basic characteristics with signs of being amphoteric.

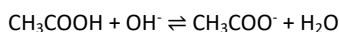
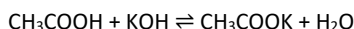
Salts of As^{+3} hydrolyze fully with formation of $\text{As(OH)}_3 \downarrow$. In character, As(OH)_3 is amphoteric with acquisition of acidic properties, Sb(OH)_3 is amphoteric with acquisition of basic properties, and Bi(OH)_3 is basic with signs of being amphoteric.

Salts of As(III) are salts of weak bases and undergo hydrolysis:



¹²Exchange reactions in electrolyte solutions proceed in the direction of combination of ions, leading to the formation of sparingly soluble substances (precipitates or gases) or of molecules of weak electrolytes. E.g., $\text{BaCl}_2 + \text{H}_2\text{SO}_4 = \text{BaSO}_4 + \text{HCl} \Rightarrow \text{Ba}^{2+} + \text{SO}_4^{2-} = \text{BaSO}_4 \downarrow$.

When both reactants and products contain sparingly soluble substances (or weak electrolytes), equilibrium shifts in the direction of formation of least soluble or least dissociated substance. E.g., neutralization of weak acid with strong base:



Two weak electrolytes – CH_3COOH and H_2O . Equilibrium shifts in the direction of formation of the weaker electrolyte – H_2O .

Also, $\text{Zn(OH)}_2 + 2\text{HNO}_3 \rightleftharpoons \text{Zn(NO}_3)_2 + 2\text{H}_2\text{O}$

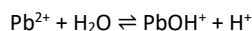


Therefore, neutralization reactions in which weak acids or bases participate are reversible. This signifies that when we dissolve in H_2O a salt comprising an anion of a weak acid or a cation of a weak base, a hydrolysis process occurs – an exchange reaction between a salt and water, the result of which is the formation of a weak acid or a weak base.

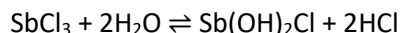
Salts from a strong acid and a strong base do not become hydrolyzed because neutralization reaction is irreversible – goes to completion.

Basic salt (Lead chloride hydroxide)

Hydrolysis of a **very** weak base: $\text{PbCl}_2 + \text{H}_2\text{O} \rightleftharpoons \text{Pb(OH)Cl} + \text{HCl}$



Salts of Sb(III) and Bi(III) are salts of **very** weak bases and are very easily hydrolyzed, forming basic salts:



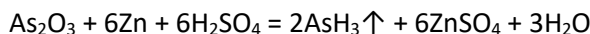
The salt that is formed, $\text{Sb(OH)}_2\text{Cl}$, is unstable and decomposes with the release of water:



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In the salt SbOCl , the group SbO plays the role of monovalent metal; this group is called antimonyl. The obtained basic salt is called antimonyl chloride or antimony oxychloride. It is basic because it has the ability to react with acids to form neutral salts: $\text{BiOCl} + 2\text{HCl} = \text{BiCl}_3 + \text{H}_2\text{O}$

Arsenic hydride or Arsine, AsH_3 , is a colourless poisonous gas with a characteristic garlic smell, less soluble in water. Arsine is formed when any compound of arsenic is reduced (in the presence of H^+ ion):



Arsine is comparatively unstable and when heated it easily decomposes to hydrogen and free arsenic. This property of arsine is used for discovering arsenic in different substances. The substance to be analyzed is acted upon by a reducer in the presence of H^+ ions, and if in it there is any compound of arsenic or arsenic in the free state, then arsine, AsH_3 , is formed. Further, the product of reduction is heated, arsine decomposes, and the released arsenic forms on the cold parts of the apparatus with a characteristic black shiny coating (deposit) called **arsenic mirror**.

Stabine or Antimony hydride, SbH_3 , is a poisonous gas and is formed in the same way as arsine. When heated, it easily decomposes to antimony and hydrogen, easier than arsine. In this case, we have formation of antimoniac mirror.

Bismuthine or Bismuth hydride, BiH_3 , is very unstable and decomposes already at room temperature.

In the series from As to Bi, stability of their compounds increases if the oxidation state of these elements is, +3. E.g., Bi in, +3, state is more stable than As in, +3, state: NaBiO_3 is a strong oxidizer while AsH_3 display reducing properties.

Arsenic acid H_3AsO_4 – in a free state is a solid substance, in aqueous solution it dissociates but less than phosphoric acid. Just like phosphoric, arsenic acid exist in several forms (ortho-, meta-). Its salts – arsenates – are derivatives of the ortho- form (type Na_3AsO_4), and just like phosphates, a majority of arsenates poorly dissolve in water.

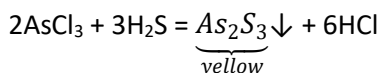
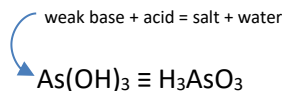
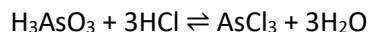
Antimonic acid H_3SbO_4 is not soluble in water and has an undefined composition: $\text{Sb}_2\text{O}_5 \cdot n\text{H}_2\text{O}$. Most of its salts – stibates – poorly dissolve in water.

Because of the high instability of Bismuth at oxidation state, +5, to obtain Bi_2O_5 is very difficult and its corresponding acid has not been obtained. However, its salts with very active metals have been obtained – NaBiO_3 .

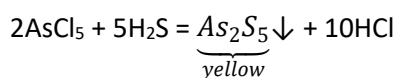
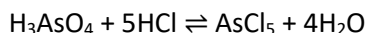
The reduction in the stability of elements in the maximum oxidation state of, +5, in the series As – Sb – Bi is displayed in the increase of oxidizing capability from compounds of As(V) to compounds of Bi(V).

Sulphides of arsenic. If hydrogen sulphide is passed through a solution of arsenous acid, acidified by HCl , then a yellow precipitate of arsenic(III) sulphide, As_2S_3 , is formed, insoluble in hydrochloric acid. The reactions taking place can be expressed by equations:

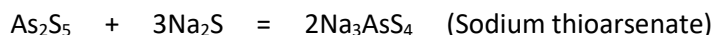
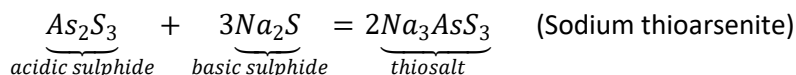
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Similarly, a yellow precipitate of arsenic(V) sulphide, As_2S_5 , can be obtained by acting hydrogen sulphide on a solution of arsenic acid in the presence of hydrochloric acid:



When sulphides of arsenic interact with sulphides of alkali metals Na_2S , K_2S or ammonium sulphide $(\text{NH}_4)_2\text{S}$, salts of thioarsenous (H_3AsS_3) and thioarsenic (H_3AsS_4) acids, which are insoluble in water, are formed. H_3AsS_3 and H_3AsS_4 acids can be seen as corresponding to oxygen-containing acids of arsenic, in which all oxygen has been replaced by Sulphur:



Sulphides of Antimony Sb_2S_3 and Sb_2S_5 are analogic in properties to sulphides of arsenic. They dissolve in sulphides of alkali metals and ammonia, forming thiosalts.

Bismuth sulphide, Bi_2S_3 , is formed as a black brownish precipitate when hydrogen sulphide acts on solutions of bismuth salts. The precipitate does not dissolve in sulphides of alkali metals and ammonia: unlike arsenic and antimony, bismuth does not form thiosalts.

The action of very strong oxidizers on compounds of bismuth(III) produces compounds of bismuth(V). The most important are bismuthates – salts of (unseparated) bismuthic acid (HBiO_3) in a free state, e.g., potassium bismuthate KBiO_3 – it is a very strong oxidizer.

Compounds of these elements (As, Sb and Bi), especially As, are more poisonous if the element is in oxidation state of, +3. Symptoms of poisoning – metallic taste in the mouth, vomiting and stomach pain. The most accessible antidote – milk, its protein – casein – forms with As insoluble compound which cannot penetrate into blood.

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Group IVA elements

Elements of group IVA have electronic formula ns^2np^2 .

Carbon and Silicon are nonmetals, Germanium, Tin and Lead are metals.

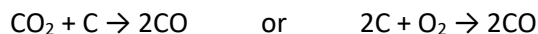
Characteristic oxidation states are, +4, +2, -4 and valency of 4. In an excited state, atoms have configuration ns^1np^3 , in this state the characteristic hybridization – sp^3 .

Carbon and its inorganic compounds

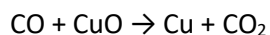
Carbon – main element from which the earth's biosphere is made. The more complex an organism is, the higher the content of carbon in it (from 0.1 to 30%). In air, composition of CO_2 is 0.03%. The main mass of carbon is in the earth's crust in form of minerals $CaCO_3$ (limestone) and $MgCa(CO_3)_2$ (dolomite).

In a free state, Carbon is found in nature in the form of diamond and graphite. There are four known modifications of carbon: two natural – diamond and graphite, two artificial – carbene and polycumulene.

Carbon(II) oxide is a neutral oxide, just like SiO, NO, N_2O and others. It can be produced:

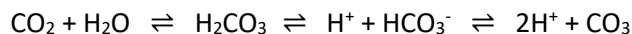


CO is a strong reducer: $3CO + Fe_2O_3 \rightarrow 2Fe + 3CO_2$



CO is poisonous, reacts with hemoglobin, forming carboxyhemoglobin.

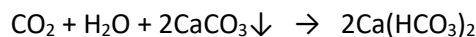
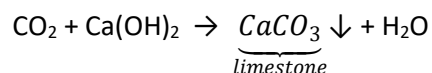
Carbon(IV) oxide interacts with water, forming weak carbonic acid H_2CO_3 :



Therefore, carbon dioxide is an anhydride of carbonic acid, i.e., carbonic anhydride.

The equilibrium is strongly shifted to the left: $CO_2 + H_2O \rightleftharpoons H_2CO_3$ Only a very small quantity of dissolved CO_2 changes to carbonic acid and can only exist in aqueous solution. When the solution is heated, CO_2 escapes, equilibrium shifts to left, and at the end, pure water remains.

CO_2 interacts with bases forming carbonates or hydro carbonates:

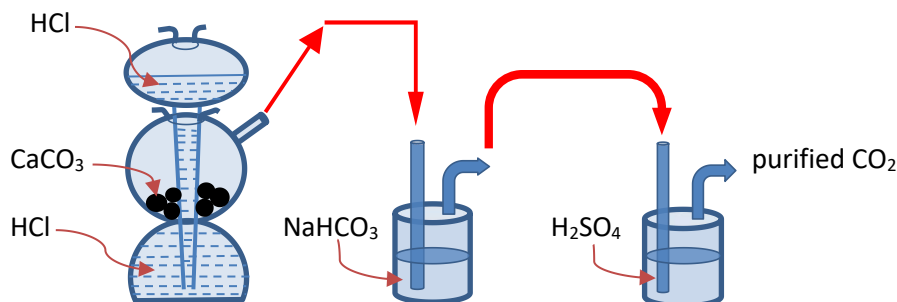


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In industry, CO_2 is produced when limestone is heated (production of unslaked lime):

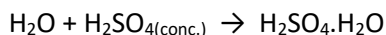
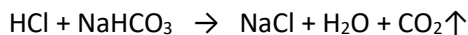


In the laboratory, CO_2 is produced in **Kipps apparatus** (diagram), by action of HCl on marble:



Kipps apparatus

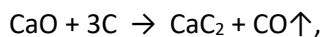
The obtained CO_2 is purified from impurities of HCl and water vapour by passing it through a solution of sodium hydrocarbonate and concentrated H_2SO_4 :



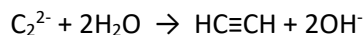
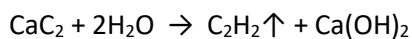
CO_2 does not burn and does not support burning, it is used for extinguishing fire. CaCO_3 is used for liming strongly acidic soils ($\text{CaCO}_3 + \text{HCl} \rightarrow \text{CaCl}_2 + \text{H}_2\text{O} + \text{CO}_2\uparrow$).

In compounds with metals, carbon displays negative oxidation states: $\overset{+3}{\text{Al}}\overset{-4}{\text{C}}_3$, $\overset{+2}{\text{Ca}}\overset{-2}{\text{C}}_2$.

Calcium carbide, which is obtained at high temperature by melting calcium oxide with carbon:



on interacting with water forms a hot gas – **acetylene**:



Also, $\text{Al}_4\text{C}_3 + 12\text{H}_2\text{O} \rightarrow 4\text{Al(OH)}_3 + 3\text{CH}_4\uparrow$

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Carbides which form acetylene when destroyed by H_2O , are called **acetylenides**. In the composition is acetylenide-ion, C_2^{2-} , which reacts with water to form $\text{HC}\equiv\text{CH}$ (reaction above). Other acetylenides are Cu_2C_2 , Ag_2C_2 and Au_2C_2 .

Carbides are crystalline bodies. The nature of the chemical bond in them can be different.

Many metal carbides of main subgroups I, II and III of the periodic table of elements are salt-forming compounds with predominant ionic bond. For example, Al_4C_3 and CaC_2 . The first one is seen as a product of hydrogen replacement by metal in CH_4 , and the second one – in C_2H_2 (C_{22}Ca_2).

In terms of chemical bonds, carbides are divided into four groups: 1) ionic or salt-forming; 2) metallic; 3) covalent; 4) carbides of cementite Fe_3C type¹³.

¹³Cementite – a chemical compound that is a constituent of steel and cast iron; very hard and brittle – iron carbide, Fe_3C .

Silicon, Germanium and their compounds

In terms of how widespread on earth elements are, Silicon occupies the second place after oxygen. Despite its chemical inertness, silicon exist in the earth only in oxygen-containing compounds.

Germanium – very scattered element, it is rare to find minerals and rocks that contain more than 1% of this element.

Just like carbon, silicon and germanium exist in the form of several modifications. Diamond-like modification of these elements are hard and shiny, infusible substances of grey-steel colour (for Si) and from silver to black colour (for Ge). Graphite-like modification of Si is not stable.

Careful baking (calcination/hardening/roasting) of the gel of silicic acid, H_2SiO_3 , produces **silica gel** – thinly dispersing powder of SiO_2 , which is used as an adsorbent.

Ability of Si atoms to form strong hetero-atomic polymeric chains, layers and lattices, explains how widespread traditional silicate material are.

Normal glass is obtained by melting a mixture of quartz¹⁴ sand with lime, CaCO_3 , and soda, Na_2CO_3 (or Na_2SO_4):
$$6\text{SiO}_2 + \text{CaCO}_3 + \text{Na}_2\text{CO}_3 = \underbrace{\text{Na}_2\text{O} \cdot \text{CaO} \cdot 6\text{SiO}_2}_{\text{mixture of Na}_2\text{SiO}_3 \text{ and CaSiO}_3 \text{ with SiO}_2} + 2\text{CO}_2\uparrow$$

Reactions that take place: $\text{CaCO}_3 + \text{SiO}_2 = \text{CaSiO}_3 + \text{CO}_2\uparrow$ // $\text{Na}_2\text{CO}_3 + \text{SiO}_2 = \text{Na}_2\text{SiO}_3 + \text{CO}_2\uparrow$

Therefore, heating as mixture of silicates (CaSiO_3 and Na_2SiO_3) with SiO_2 , $\text{Na}_2\text{O} \cdot \text{CaO} \cdot \text{SiO}_2$ is obtained.

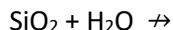
¹⁴Quartz – a hard glossy mineral consisting of SiO_2 in crystal form; present in most rocks (especially sandstone and granite). Yellow sand is quartz with FeO impurities. Colourless glass is made of almost pure silica; quartz glass (SiO_2).

Silicon is obtained by reaction: $\text{SiO}_2 + 2\text{Mg} \rightarrow \text{Si} + 2\text{MgO}$ // $\text{SiO}_2 + 2\text{C} \rightarrow \text{Si} + 2\text{CO}\uparrow$

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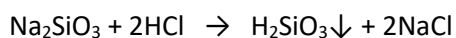
Silicon is stable to acids, but soluble in bases: $\text{Si} + 2\text{NaOH} + \text{H}_2\text{O} \rightarrow 2\text{H}_2\uparrow + \text{Na}_2\text{SiO}_3$

Silicon(IV) oxide is the most stable compound of silicon. It is found in crystalline as well as in amorphous form. SiO_2 is an acidic oxide, but, having polymeric structure, it does not dissolve in water:



However, long continuous exposure of normal glass to water causes water to partly extract sodium silicate from glass. For example, if grinded glass powder is put in a container with water and the container thoroughly shaken, then, when a few drops of phenolphthalein are added to the contents of the container, the liquid turns pink – hydrolysis of sodium silicate.

Silicic acid, very weak, can be obtained by the action of acids on its salts (silicates):

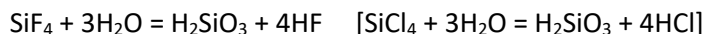


Silicic acid is not soluble in water and when strongly heated, decomposes to silicon dioxide and water.

Silicates of sodium and potassium are soluble in water, silicates of other metals are not soluble.

SiF_4 is formed when $\underbrace{\text{SiO}_2}_{\text{sand}} + 4\text{HF} = \text{SiF}_4\uparrow + 2\text{H}_2\text{O}$. It is a colourless gas with a pungent smell.

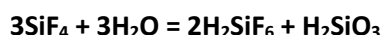
Just like SiCl_4 , SiF_4 undergoes hydrolysis in aqueous solution:



The obtained HF interacts with SiF_4 . As a result **fluosilicic** acid, H_2SiF_6 , is produced:



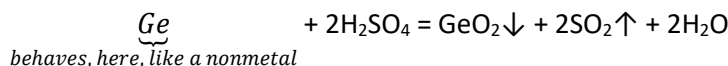
The total process can be expressed by equation:



In strength, fluosilicic acid approaches sulphuric acid.

At room temperature, Germanium is stable to action of air, oxygen, water, HCl and dilute H_2SO_4 .

HNO_3 and conc. H_2SO_4 oxidize Ge up to GeO_2 , especially on heating:



Ge also interacts with alkalis in the presence of H_2O_2 . As a result, salts of Germanic acid, H_2GeO_3 , are formed: $\text{Ge} + 2\text{NaOH} + 2\text{H}_2\text{O}_2 = \text{Na}_2\text{GeO}_3 + 3\text{H}_2\text{O}$ [compare this to $\text{Si} \rightarrow \text{salt of acid, Na}_2\text{SiO}_3 + \text{H}_2$]

Compounds of germanium(II) are less stable. The most characteristic oxidation state for germanium in compounds is, +4.

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Germanium has semi-conducting properties – and hence, its main use.

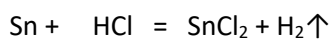
Tin, Lead and their compounds

In a free state, Tin is a silver-white soft metal. Two types exist – white tin (stable at $t^\circ > 14^\circ\text{C}$) and grey tin (stable at $t^\circ < 14^\circ\text{C}$). On cooling, white tin changes into grey – significant change in density occurs – the metal becomes grey powder (tin pest).

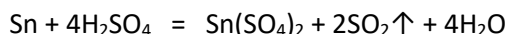
In air, tin does not oxidize, but when heated beyond melting point, it slowly changes to SnO .

Water does not react on tin.

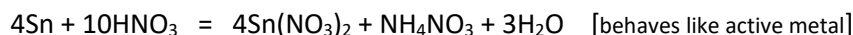
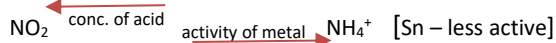
Diluted HCl and H_2SO_4 react very slowly on tin, which is why it takes long for H_2 to evolve:



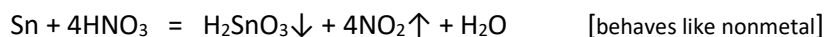
Concentrated solutions of HCl and H_2SO_4 dissolve tin, especially on heating:



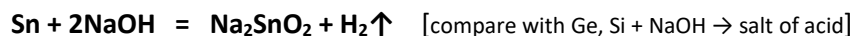
With HNO_3 , tin interacts intensively the higher the concentration and temperature. In **dilute** acid, soluble tin(II) nitrate is formed:



In **concentrated** acid, a compound of tin(IV), mainly insoluble β -stannic acid is formed:



Concentrated bases also dissolve tin. In this case, stannites are obtained – salts of stannous acid, H_2SnO_2 :



In air, tin is covered by a thin oxide film which has a protective function. Hence, it is a chemically stable metal. It is used for conserving food – the product of corrosion is harmless.

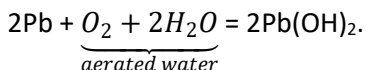
Tin has stable compounds, in which oxidation state is, +2 and +4.

The most important ore from which Lead is mined is lead glance, PbS .

Lead is a bluish-white heavy metal. It is very soft and is easily sliced with a knife.

In air, lead is quickly blanketed by a thin layer of oxide, which protects it from further oxidation.

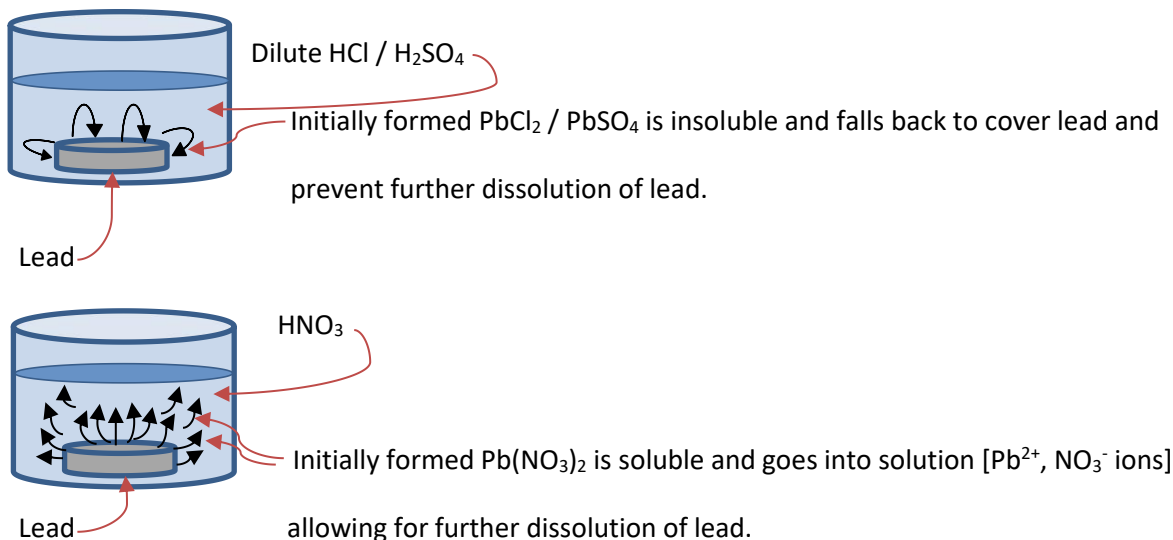
Water by itself does not react with lead, but in the presence of air, lead is slowly destroyed by water with the formation of lead(II) hydroxide:



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However, when in contact with hard water, lead coats a protective layer of insoluble salts (sulphates, lead carbonates) which prevents further action by water and formation of hydroxides.

Diluted HCl and H_2SO_4 practically do not react with lead. This is connected with significant restraint (difficulty) of evolving hydrogen, and also with the poor solubility of PbCl_2 and PbSO_4 which then cover the surface of the dissolving metal (diagram).



In HNO_3 , lead easily dissolves, in acid of less concentration it dissolves much quicker than in acid of higher concentration. This is because solubility of the product of corrosion (lead nitrate) decreases with increase in concentration of acid [high concentration means less water (by %), therefore, lead dissolves less: $\text{Pb}(\text{NO}_3)_2$ is in aqueous solution].

Likewise, lead easily dissolves in acetic acid that contains dissolved oxygen.

Lead is also soluble in bases, although at a slower pace; intensity increases with hot dilute solutions. As a result, hydroxoplumbites are formed: $\text{Pb} + 4\text{KOH} + 2\text{H}_2\text{O} = \text{K}_4[\text{Pb}(\text{OH})_6] + \text{H}_2\uparrow$

All soluble compounds of lead are poisonous.

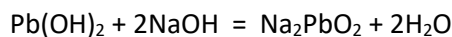
Characteristic oxidation state for lead is, +2 and, +4. The **most stable** oxidation state of lead in most compounds is, +2.

Lead monoxide, PbO , is a yellow powder, formed after heating molten lead in air. After roasting at temperature 500°C , it acquires a reddish yellow colour and in this form it is called **litharge**. PbO is used for filling cells in storage batteries.

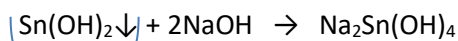
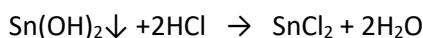
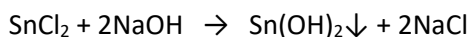
Lead(II) hydroxide, $\text{Pb}(\text{OH})_2$, is formed when alkali acts on soluble salts of lead(II). It has amphoteric properties and dissolves in acids forming salts of lead(II), and in alkalis with the formation of hydroxoplumbites: $\text{Pb}(\text{OH})_2 + 4\text{NaOH} = \text{Na}_4[\text{Pb}(\text{OH})_6]$

On melting $\text{Pb}(\text{OH})_2$ with dry alkalis, salts called plumbites are obtained:

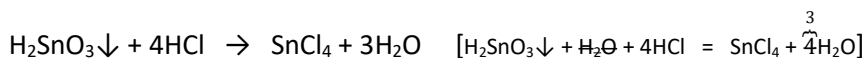
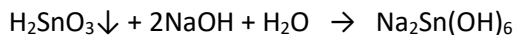
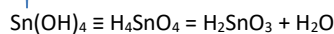
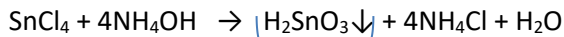
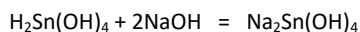
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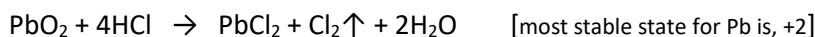
Tin and lead hydroxides have amphoteric properties. In this case, basic properties are displayed in oxidation state, +2, and, in oxidation state, +4, - acidic properties.
for example Sn(OH)_2
e.g., H_2SnO_3



Acting as an acid, but the expected product is hydroxo complex: $\text{H}_2\text{SnO}_2 \rightleftharpoons \text{H}_2\text{Sn(OH)}_4$
 H_2SnO_2 has a charge of -2, and $\text{H}_2\text{Sn(OH)}_4$ has a charge of -2.



Compounds of Sn^{2+} have reducing properties, and compounds of Pb^{4+} - oxidizing.



Group IIIA elements

Elements of group IIIA have electronic formula ns^2np^1 . They are considerably less active than alkaline-earth metals. Their characteristic oxidation state is, +3, and valency of 3. During formation of covalent bond, $s \rightarrow p$ electron excitement and sp^2 hybridization of AOs take place.

Metallic properties of elements in the group increases from up downwards, reducing properties of their atoms grow. The basic properties of their hydroxides increase and their acidic properties reduce.

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B – nonmetal,	B ₂ O ₃ – acidic,	H ₃ BO ₃ – acid
Al – metal,	Al ₂ O ₃ – amphoteric,	Al(OH) ₃ – amphoteric hydroxide
Ga – metal,	Ga ₂ O ₃ – amphoteric,	Ga(OH) ₃ – amphoteric hydroxide
In – metal,	In ₂ O ₃ – amphoteric,	In(OH) ₃ – amphoteric hydroxide
Tl – metal,	Tl ₂ O ₃ – basic,	Tl(OH) ₃ – base

Compounds of Tl³⁺ are strong oxidizers and are reduced to Tl⁺.

Boron and its compounds

Natural Boron comprises 80% of isotope ¹¹B and 20% of isotope ¹⁰B. Boron is found in nature in form of compounds, e.g., borax Na₂B₄O₇·10H₂O.

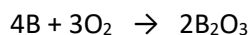
Crystalline Boron is greyish-black, semiconductor, in hardness it approaches diamond, and it is infusible.

With respect to its own chemical properties, boron resembles silicon, in which the rule of “diagonal affinity”¹⁵ in the periodic system of elements is displayed.

¹⁵Similarities along diagonals in the periodic table. E.g., electronegativity values along diagonals are quite similar (from B to Te: the values range between 1.9 and 2.2).

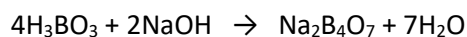
In normal conditions Boron displays chemical inertness, reacting only with fluorine to change into BF₃. At high temperatures, it is oxidized by oxygen (in B₂O₃), Sulphur (in B₂S₃), chlorine (in BCl₃), and nitrogen (in BN). Hydrogen does not interact directly with Boron. Boron resists the action of alkalis and acids, with the exception of “aqua regia” and hot concentrated solutions of HNO₃ and H₂SO₄, which changes boron to H₃BO₃.

Boron oxide is an acidic oxide and dissolves in water with the formation of weak ortho-boric acid:



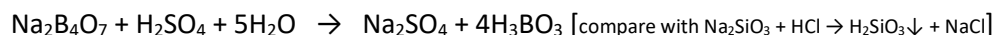
Orthoboric acid H₃BO₃, when heated, decomposes to metaboric acid HBO₂, tetraboric acid H₂B₄O₇ and boric anhydride B₂O₃: $\text{H}_3\text{BO}_3 \xrightarrow{t} \text{HBO}_2 \xrightarrow{t} \text{H}_2\text{B}_4\text{O}_7 \xrightarrow{t} \text{B}_2\text{O}_3$

Orthoboric acid, H₃BO₃, interacts with alkalis and forms salts of tetraboric acid:

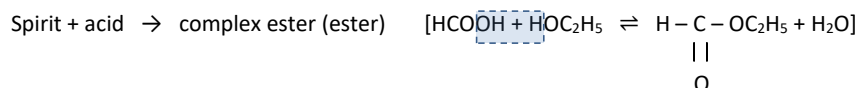
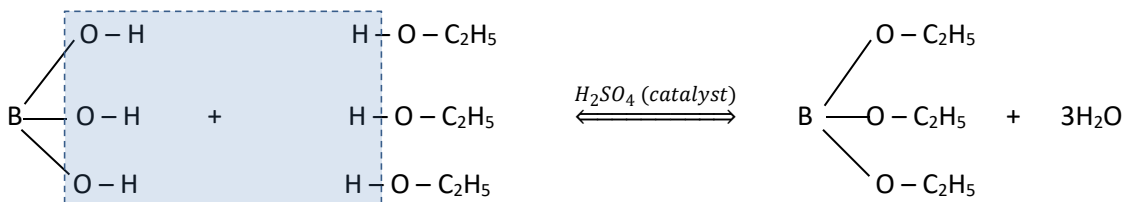


When strong acids interact with sodium tetraborate, weak orthoboric acid is forced out:

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Orthoboric acid interacts with ethanol in the presence of concentrated H_2SO_4 with the formation of ethyl borate, which burns with a green flame:¹⁶



Reaction of formation of complex ester from acid and spirit (or phenol) is called esterification. It is catalyzed by H^+ ions and, therefore, is accelerated in the presence of mineral acids. Equilibrium is reached quickly in the presence of mineral acid since ions of hydrogen not only accelerates the process of esterification, but, also, the reverse process of hydrolysis.

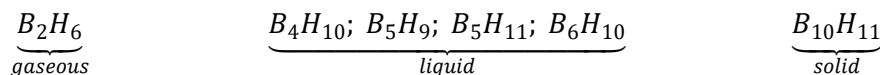
Boron is a microelement. Microelements (or micro fertilizers) are elements B, Cu, Mn, Zn, Co, Mo and others. They increase the activity of enzymes, catalyzes biochemical processes, promote the synthesis of proteins and nucleic acid, vitamins, sugars and starch, influences photosynthesis, and accelerates growth of plants and ripening of seeds. Compounds of boron influences carbohydrate exchange and are needed for normal growth and formation of seeds.

In alloying with metals, boron acts as an oxidizer, exhibiting oxidation state of, -3, in compounds with s-elements (e.g., magnesium boride, Mg_3B_2). These compounds, which are called borides, are decomposed by acids with the formation of a mixture of boron hydrides, called boranes:



Boranes are produced either from borides (type Mg_3B_2) by decomposing them by acids or from boron halides (BF_3 and BCl_3) by acting hydrogen on them ($\text{BF}_3 + \text{H}_2 \rightarrow \text{B}_n\text{H}_n + \dots$) or acting substances which contain hydrogen as an anion, e.g., NaH , NaBH_4 ($\text{BF}_3 + \text{NaH} \rightarrow \text{NaF} + \text{B}_n\text{H}_n + \dots$).

About 30 boranes are known. They can be divided into two groups: B_nH_{n+4} and B_nH_{n+6} .



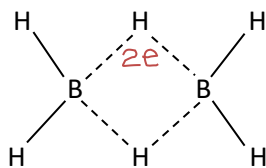
Nomenclature for boranes: the number of B atoms is specified by a prefix and the number of H atoms is given in parentheses. E.g., B_2H_6 – diborane (6), B_4H_{10} – tetraborane (10)

The main product of the interaction of Mg_3B_2 with HCl is tetraborane, B_4H_{10} , which is a volatile liquid with very unpleasant smell and which ignites in air. When stored, it slowly decomposes to form

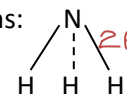
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diborane, B_2H_6 . B_2H_6 does not burn in air, and in water it decomposes giving out hydrogen and forming boric acid, H_3BO_3 : $B_2H_6 + 6H_2O = 2H_3BO_3 + 6H_2$

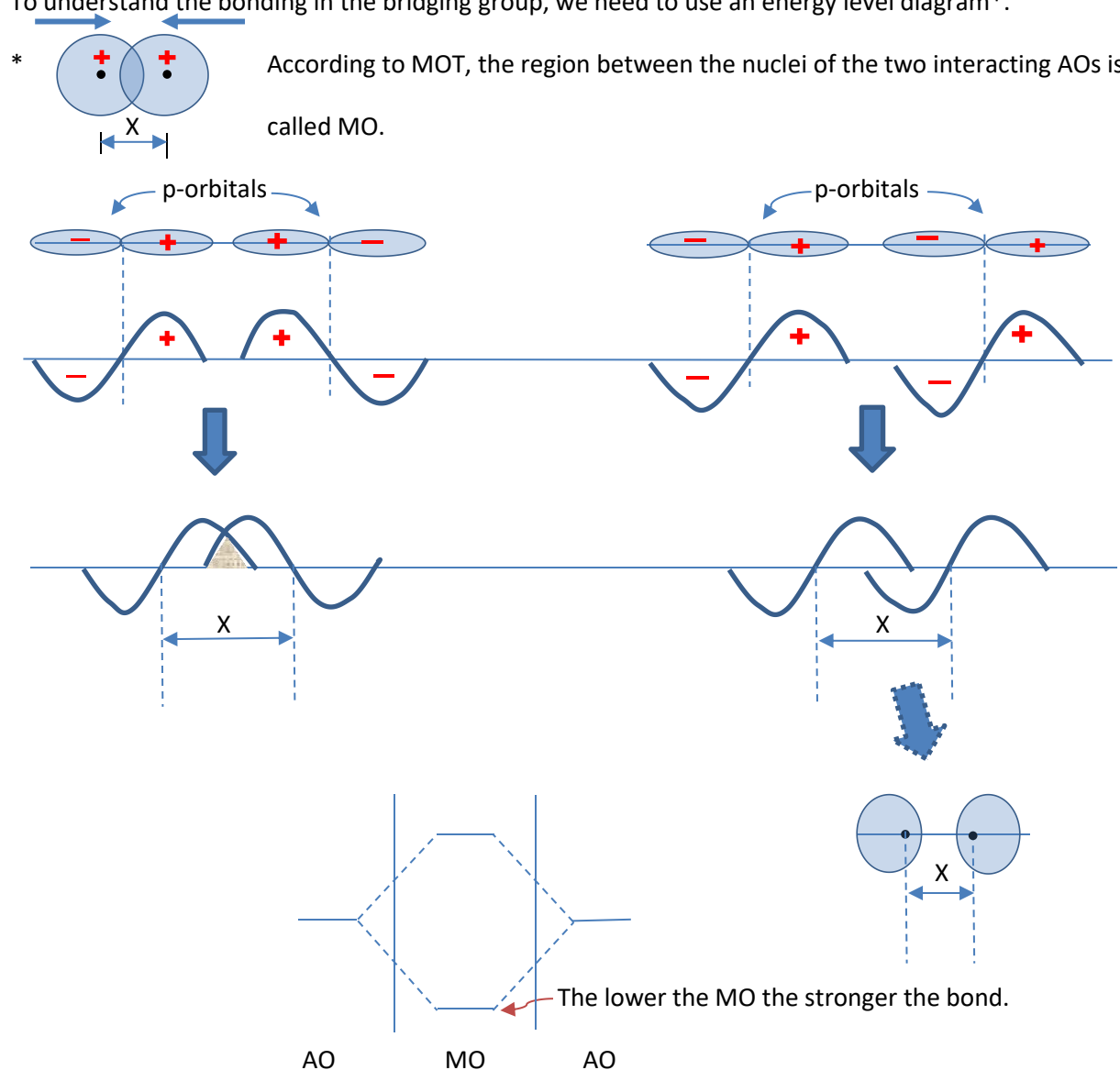
Let's look at the structure of B_2H_6 :



The bridging bond B-H-B has three atoms participating and two electrons bonding the three atoms. But for example in NH_3 , there are two atoms participating for each bond and two electrons:



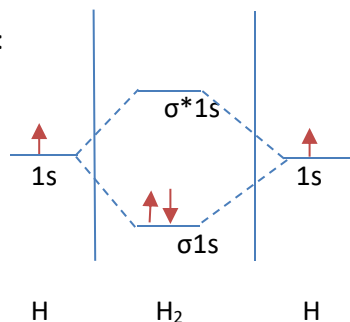
To understand the bonding in the bridging group, we need to use an energy level diagram*.



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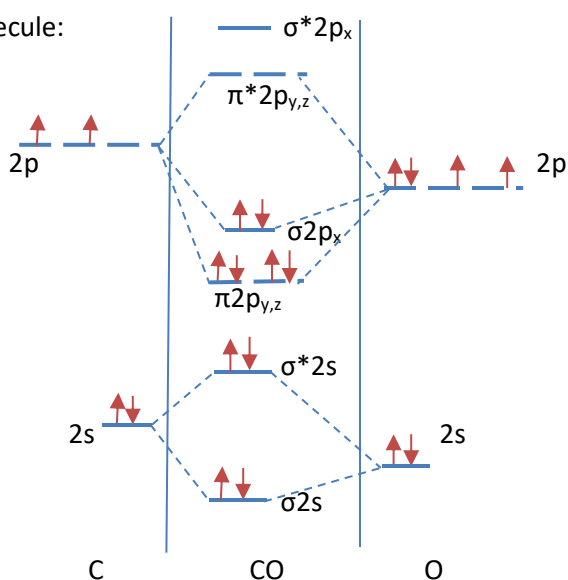
$\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \sigma 2p_x < \pi 2p_{y,z} < \pi^* 2p_{y,z} < \sigma^* 2p_x$ – energy of MOs. Rearranging to give correct arrangement for period 2 elements, we get: $\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \pi 2p_{y,z} < \sigma 2p_x < \pi^* 2p_{y,z} < \sigma^* 2p_x$.

H₂ molecule:



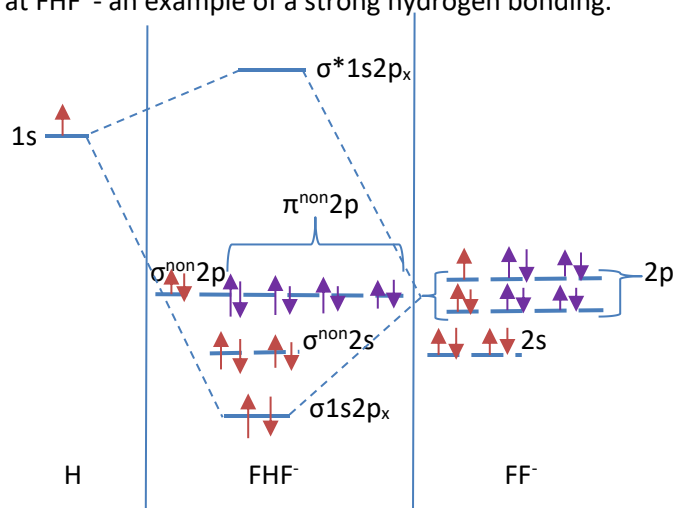
Here, two electrons are in bonding MO. Two electrons bond two atoms.

CO₂ molecule:



Bonding electrons are in bonding MOs.

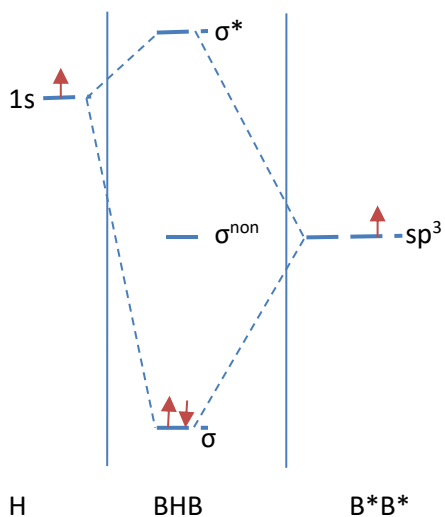
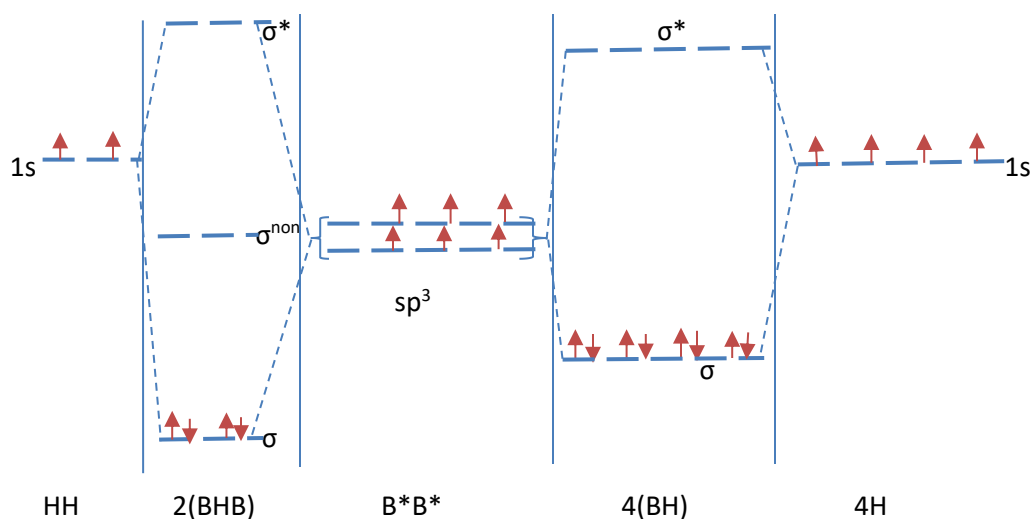
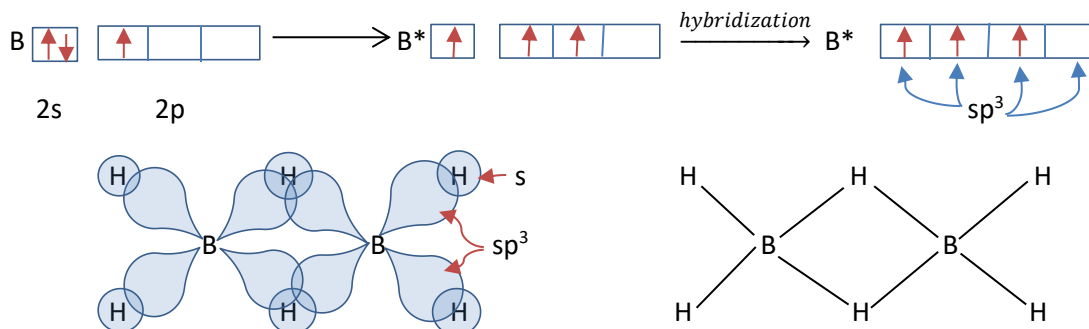
Let's look at FHF⁻ - an example of a strong hydrogen bonding.



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This is a 3-centre, 2-electron bond (3c, 2e bond), i.e., it has a single MO containing a pair of bonding electrons delocalized over all three atoms.

In boron hydrides, called **boranes**, hydrogen often serves as a bridge:

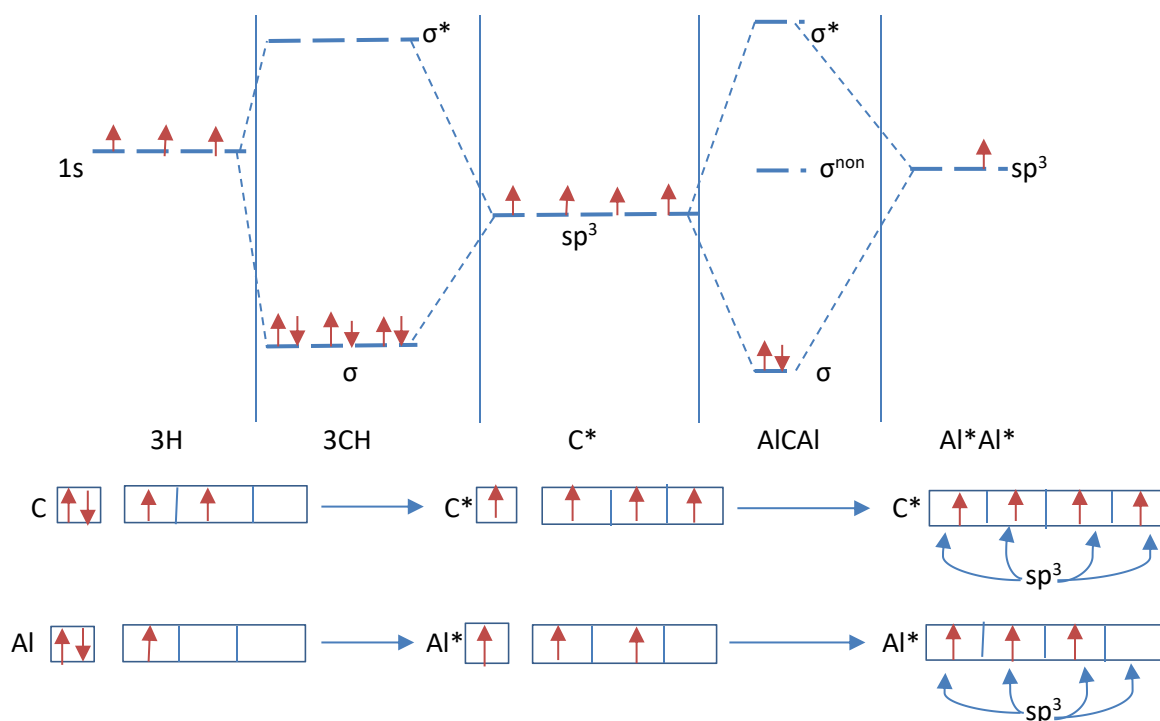
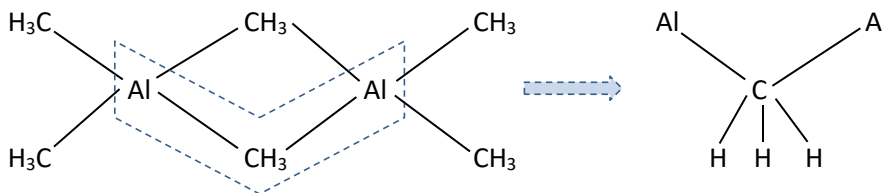


From the bridging BHB group, there is formation of three MOs, one bonding, one nonbonding and one antibonding, from the two sp³ HOs of the boron atoms and the 1s AO of hydrogen atom.

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Two electrons (half of available four) are available to occupy the bonding MO. Stabilization of these electrons by formation of the bond holds these atoms together. An identical set of interactions accounts for the bonding in the other B-H-B bridge. This type of bonding, involving three atoms and two bonding electrons, is described as 3-centre, 2-electron (3c, 2e) bonding.

Similar bridging hydrogen atoms occur in many other boranes, as well as in carboranes, which contain both boron and carbon atoms arranged in clusters. In addition, bridging hydrogen and alkyl groups are frequently encountered in **aluminium** chemistry and in the chemistry of elements of other groups.



^{10}B has a very high neutron absorption cross-section (it is a good absorber of neutrons). This property is used in the treatment of cancerous tumor: boron-containing compounds having a strong preference for attraction to tumor sites can be irradiated with beams of neutrons; the subsequent nuclear decay can kill the cancerous tissue.

Let us consider the structure of the cage-like boron hydrides.

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These include the series B_nH_{n+4} and B_nH_{n+6} as well as more recently discovered $[B_nH_n]^{2-}$ **closed polyhedra**. The boron hydrides occur with a variety of shapes, some resembling nests and others butterflies.

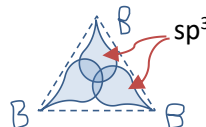
All the boron hydrides are electron deficient.

William Lipscomb and his students in the USA used single-crystal x-ray diffraction to determine the structures of a large number of boron hydrides, and extend the concept of multicenter bonding to these more complex compounds.

The boron cluster¹⁷ compounds are best considered from the standpoint of fully delocalized MOs containing electrons that contribute to the stability of the entire molecule¹⁸ or skeletal structure (frame), distributed to stabilize the entire structure. [Remember FHF^- : MO with $2e^-$ delocalized over all three atoms].

¹⁷Cluster – multinuclear compound, in which a chemical bond forms directly between central atoms, e.g., in our case, $B-B$ and for complexes, $(CO)_5Mn-Mn(CO)_5$.

In more complex boranes, the three centres of the $3c, 2e$ bonds may be $B-H-B$ bridge bonds (as in diborane), but they may also be bonds in which three B atoms lie at the corners of an equilateral triangle with their sp^3 hybrid orbitals overlapping at the centre:



¹⁸E.g., the state collects taxes from its citizens in order to build infrastructure, Medicare, etc. This in turn will help in building a prosperous society. Prosperity means stability of society (state). Not everyone contributes tax for the pool. But when redistributing, the entire society benefits.

Classification and electron counting

A correlation between the numbers of electrons (counted in a specific way), the formula, and the shape of the molecule was established by British chemist Kenneth Wade in the 1970s.

Wade's rules apply to a class of polyhedra¹⁹ called **deltrahedra** (because they are made up of triangular faces) and can be used in two ways.

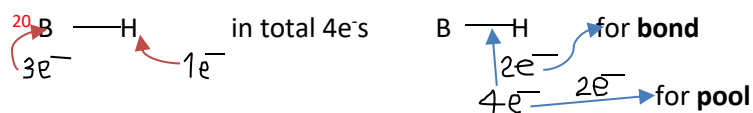
¹⁹Polyhedron – many faced object.

For boranes and borohydrides (e.g., $B_{10}H_{11}$), they enable us to predict the general shape of the molecule or anion from its formula.

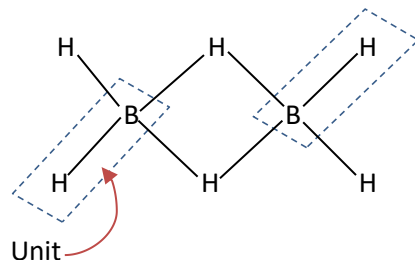
For boranes, e.g., B_2H_6 , the building block from which the deltahedron is constructed is assumed to be one $B-H$ group. [For mammals, the building block are bones which make up the skeleton (framework)].

The electrons in the $B-H$ bond are ignored²⁰ in the counting procedure, but all others are included whether or not it is obvious that they help to hold the skeleton together.

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By skeleton we mean the framework of the cluster, with each B – H group counted as a **unit**. For instance, in B_2H_6 , the two B atoms have two “terminal” H atoms each, but only one B – H bond is treated as a unit, the other pair of electrons is treated as part of the skeleton. A B – H group makes two electrons available to the skeleton (the B atom provides three and the H atom one, but of these four electrons, two are used for the B – H bond).

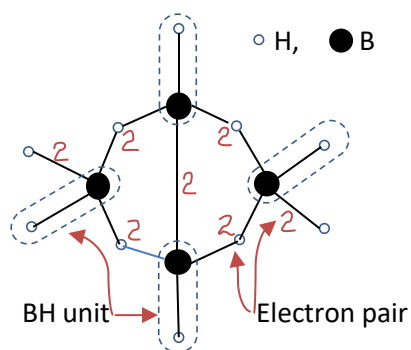


$$B - H = 2e^- \times 2 = 4e^-$$

$$4H = 4e^-$$

$$\text{total} = 8e^-$$

For example: Count the number of skeletal electrons in B_4H_{10} .



Four B – H units contribute $4 \times 2e^- = 8e^-$, and

Six additional H atoms contribute a further $6e^-$.

In total there are $14e^-$.

The resulting 7 pairs are distributed to stabilize the entire molecule as shown in the diagram:

Two are used for the additional terminal B – H bonds, four are used for the four B – H – B bridges, and one is used for the central B – B bond.

According to Wade's rules (table, below), boranes of formula $[B_nH_n]^{2-}$ will have the **closo** structure (from Greek for “cage”) with a B atom at each corner of a closed deltahedron and no B – H – B bonds. Such structures have **$n + 1$ pairs** of skeletal electrons. This series of anions is known for $n = 5$ to $n = 12$ and examples include the trigonal bipyramid $[B_5H_5]^{2-}$ ion, the octahedral $[B_6H_6]^{2-}$ ion, and the icosahedral $[B_{12}H_{12}]^{2-}$ ion.

Boranes of formula B_nH_{n+4} have the **nido** structure (from Latin for “nest”). They can be viewed as a closo-borane that has lost one vertex but may have B – H – B bonds as well as B – B bonds. The compounds in this series contain **$n + 2$ pairs** of skeletal electrons. Example is B_5H_9 . In general, the thermal stability of nido-boranes is intermediate between that of closo- and arachno- boranes.

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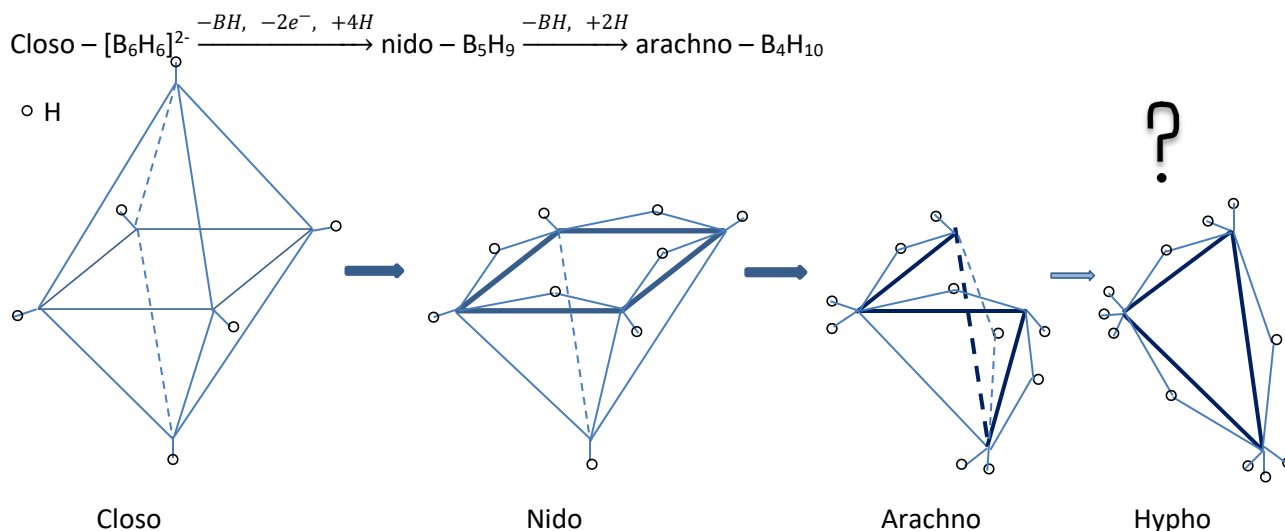
The boranes of formula B_nH_{n+6} have the **arachno** structure (from Greek for “spider”) and are closo-borane polyhedra less two vertices (and may have B – H – B bonds); they are so called because they resemble an untidy spider’s web. The arachno-boranes have **n + 3 skeletal electron pairs**. Example is B_5H_{11} .

Type	Formula	Skeletal electron pairs	Examples
Closo	$[B_nH_n]^{2-}$	$n + 1$	$[B_5H_5]^{2-}$ to $[B_{12}H_{12}]^{2-}$
Nido	B_nH_{n+4}	$n + 2$	B_2H_6 , B_5H_9 , B_6H_{10}
Arachno	B_nH_{n+6}	$n + 3$	B_4H_{10} , B_5H_{11}
Hypho (from Greek for “net”)	B_nH_{n+8}	$n + 4$	None

In some cases, protons can be removed, thus, $[B_5H_8]^-$ is a nido analogue of B_5H_9 .

A very useful structural correlation between closo, nido and arachno compounds is based on the observation that clusters having the same number of skeletal electrons are related by removal of successive B – H groups and addition of the appropriate number of electrons and H atoms.

This type of process relates the octahedral closo – $[B_6H_6]^{2-}$ anion to the square-pyramidal nido – B_5H_9 borane, and that in turn to the butterfly-like arachno – $[B_4H_{10}]$ molecule:



Exercise: Count the number of skeletal electron pairs for the closo-borane $[B_6H_6]^{2-}$.

There are six B – H units and hence 12 skeletal electrons plus two from overall, -2 charge. Total number of pairs is seven (7), or **n + 1**, skeletal electron pairs. This is characteristic of closo clusters.

The closo structures make up only a very small fraction of all known borane species. Additional structural types can be obtained by removing one or more corners from the closo framework. Removal

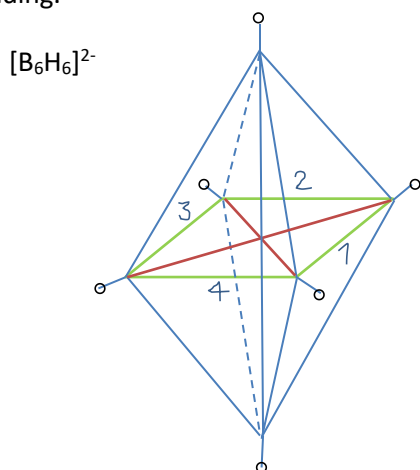
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of one corner yields a nido (nest-like) structure, removal of two corners an arachno (spider web-like) structure, and removal of three corners a hypho (net-like) structure as shown above.

Classification scheme of cluster structures based on Wade's rules:

Structure type	Pairs of framework bonding electrons	Empty corners
Closo-	$n + 1$	0
Nido-	$n + 2$	1
Arachno-	$n + 3$	2
Hypho-	$n + 4$	3

It is sometimes useful to relate the **total valence electron count** in boranes to the structural type. In closo boranes, the total number of valence electron pairs is equal to the **sum of the number of vertices** in the polyhedron (at each boron one pair is involved in boron-hydrogen bonding) and the **number of skeletal bond pairs**. For example, in $B_6H_6^{2-}$, there are 26 valence electrons, or 13 pairs. Six of these pairs are involved in bonding to the hydrogen (one per boron), and seven pairs are involved in framework bonding.



In red (three B – B bonds) – three electron pairs.

In green (1, 2, 3, 4) – four B – B bonds – 4 electron pairs.

In total 7 electron pairs for framework bonding.

The polyhedron of the **closo** structure is the **parent polyhedron** for the other structural types.

Suppose we are given the formula of a borane structure and asked to sketch the structure, then, we first need to know the parent structure (closo structure). We need to classify this borane.

Boranes can conveniently be classified by considering

Closo boranes to have the formula $B_nH_n^{2-}$;

Nido boranes to be derived from $B_nH_n^{4-}$ ions;

Arachno boranes to be derived from $B_nH_n^{6-}$ ions;

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Hypho boranes to be derived from $B_nH_n^{8-}$ ions.

The formulas of boranes can be related to these (above) formulas by formally subtracting H^+ ions. For example, to classify $B_9H_{14}^-$ we can formally consider it to be derived from:

$$B_9H_{14}^- - 5H^+ = B_9H_9^{6-} \quad [-1 - 5 = -6]$$

The appropriate classification for this borane is therefore, **arachno**.

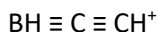
Another example: classify by structural type $B_{10}H_{14}$.

$$B_{10}H_{14} - 4H^+ = B_{10}H_{10}^{4-}, \text{ a nido borane.}$$

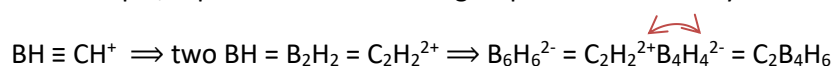
Heteroboranes

Electron counting schemes can be extended to isoelectronic²¹ species such as the carboranes (carbaboranes). The CH^+ unit is isoelectronic with BH ; many compounds are known in which one or more BH groups have been replaced by CH^+ (or by C , which has the same number of electrons as BH).

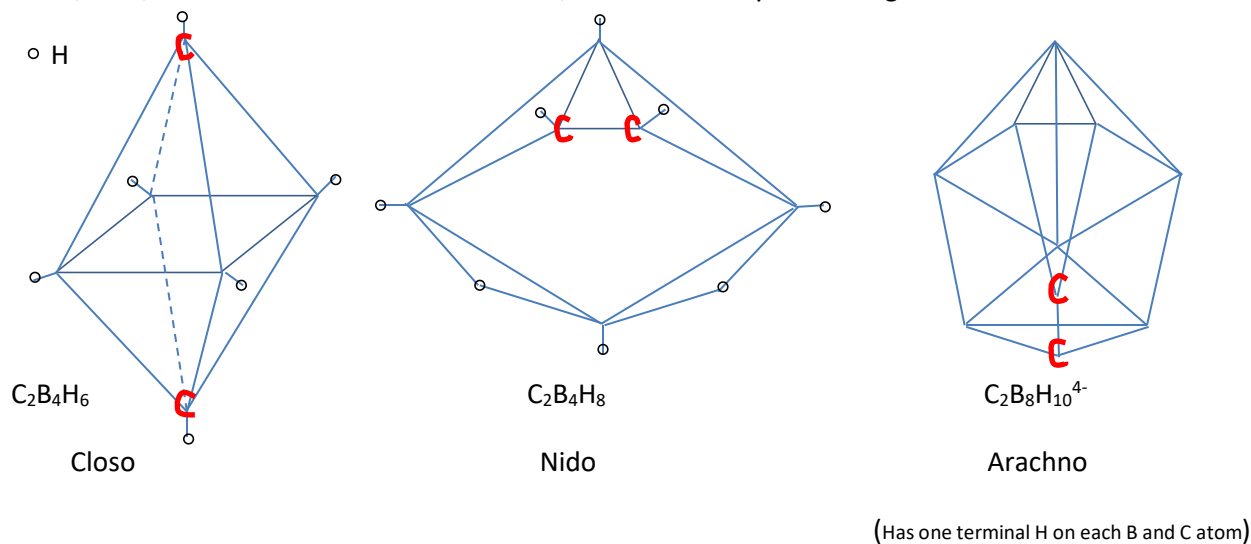
²¹No of electrons is same: $B \uparrow\downarrow \uparrow \uparrow \uparrow \uparrow$ $C \uparrow\downarrow \uparrow \uparrow \uparrow \uparrow$



For example, replacement of two BH groups in closo- $B_6H_6^{2-}$ yields closo- $C_2B_4H_6$, a neutral compound:



Closo, nido, arachno carboranes are all known, most commonly containing two carbon atoms:

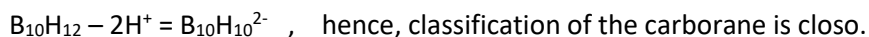
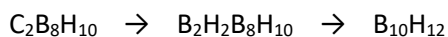


Examples of chemical formulas corresponding to these designations are:

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Type	Borane	Carborane
Closo	$B_{12}H_{12}^{2-}$	$C_2B_{10}H_{12}$ [BH $\equiv$ CH ⁺]
Nido	$B_{10}H_{14}$	$C_2B_8H_{12}$ [BH $\equiv$ C] in the case where there is no charge
Arachno	B_9H_{15}	$C_2B_7H_{13}$ [BH $\equiv$ C]

Carboranes may be classified by structural type using the same method as for boranes. Since a carbon atom has the same number of valence electrons as a boron atom plus a hydrogen atom, formally each C should be converted to BH. For example, for carborane $C_2B_8H_{10}$:



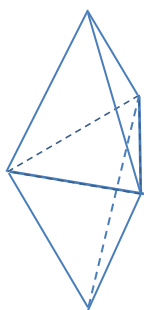
Many derivatives of boranes containing other main group atoms (heteroatom) are also known. These heteroboranes may be classified by formally converting the heteroatom to a BH_x group having the same number of valence electrons, then proceeding as in previous examples. For some of the most common heteroatoms the substitutions are:

Heteroatom	Replace with
C, Si, Ge, Sn	BH
N, P, As	BH_2 [B  N ]
S, Se	BH_3

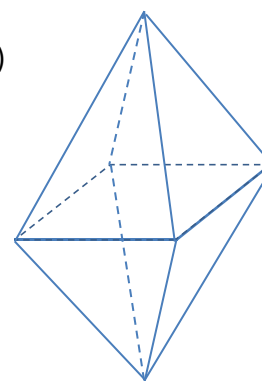
Exercise: Classify heteroboranes SB_9H_{11} and $CPB_{10}H_{11}$ by structural type.

Sketches of closo structures, from $B_5H_5^{2-}$ to $B_{12}H_{12}^{2-}$ (Deltahedra):

Trigonal bipyramid (6 faces)

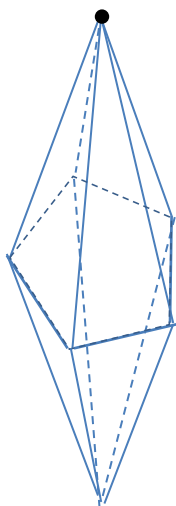


Octahedron (8 faces)

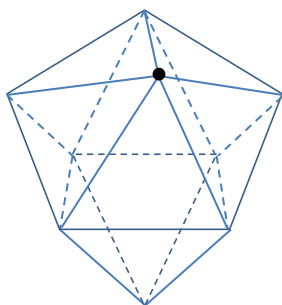


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Pentagonal bipyramid (10 faces)

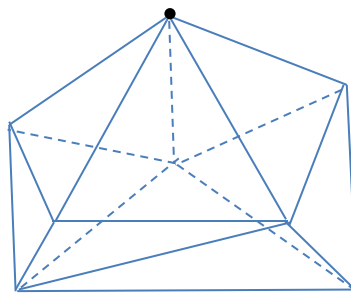


Capped square antiprism (14 faces)



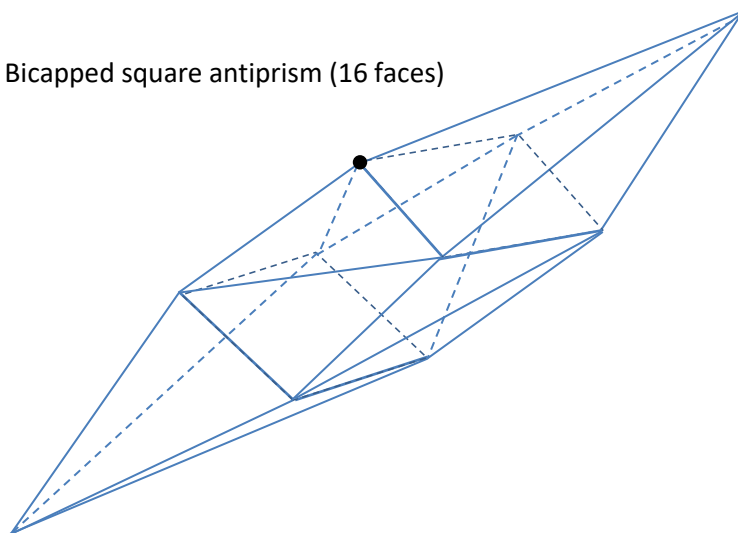
- Capping boron

Dodecahedron (12 faces)



- Capping Boron

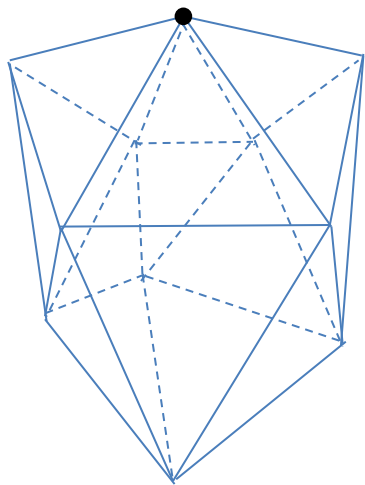
Bicapped square antiprism (16 faces)



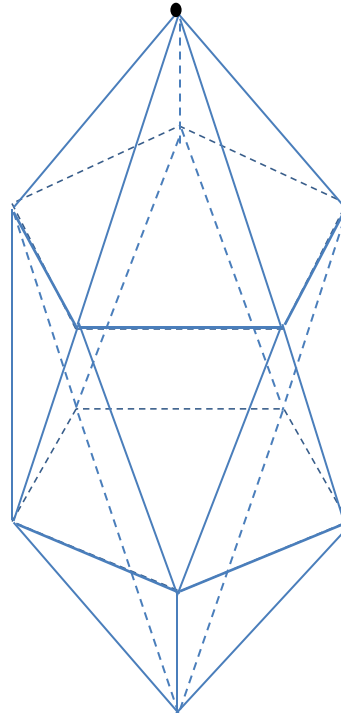
- Capping boron

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Octadecahedron (18 faces)



Icosahedron (20 faces)



Aluminium and its compounds

Aluminium is the most widespread element in the earth's crust. It is found in clay, mica²² and other minerals. Aluminium's content in earth's crust is 8% (in mass).

²²Mineral consisting of silicates of aluminium.

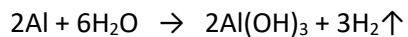
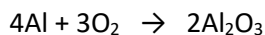
In industry, aluminium is produced from electrolysis of the fusion of bauxite, Al_2O_3 , in cryolite, Na_3AlO_3 , at 960°C .



At the cathode, reduction is taking place: $\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$

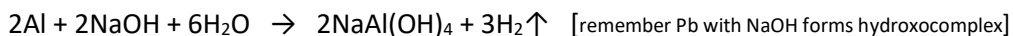
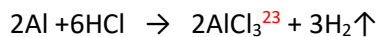
At the anode, oxidation is taking place: $2\text{AlO}_3^{3-} - 6\text{e}^- \rightarrow \frac{3}{2}\text{O}_2 + \text{Al}_2\text{O}_3$

Aluminium is covered by an inert protective oxide film, Al_2O_3 . Without the oxide film, Aluminium is very active and actively interacts with oxygen and water:

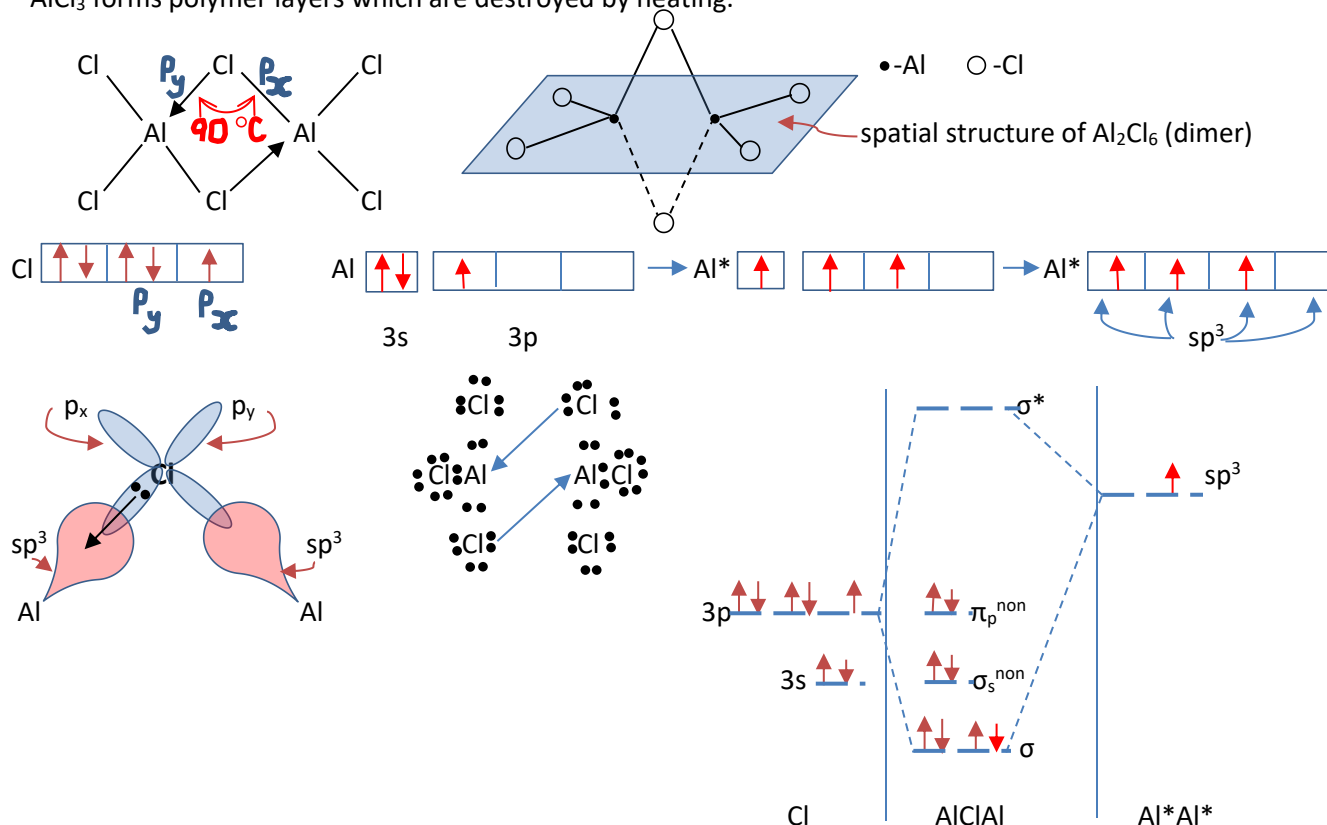


Aluminium interacts with acids and alkalis:

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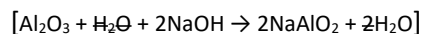
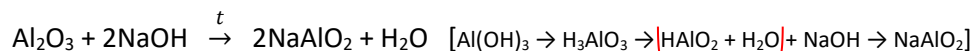
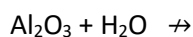
²³AlCl₃ forms polymer layers which are destroyed by heating.



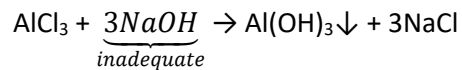
The bridging group Al-Cl-Al has a 3c, 4e bond. 4 electrons bind 3 atoms.

Compare the above energy level diagram with the one for AlH₃. AlH₃ is also electron deficient, but H has no pair of electrons to donate, but acts also as a bridging atom.

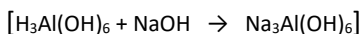
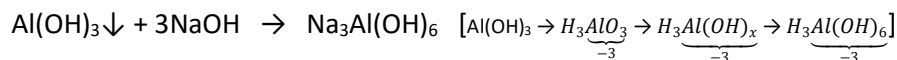
Aluminium oxide has amphoteric properties.



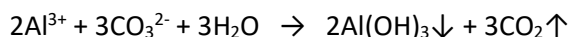
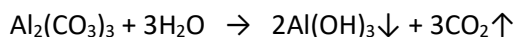
Aluminium hydroxide can be obtained from salt by the action of inadequate alkali. It has amphoteric properties:



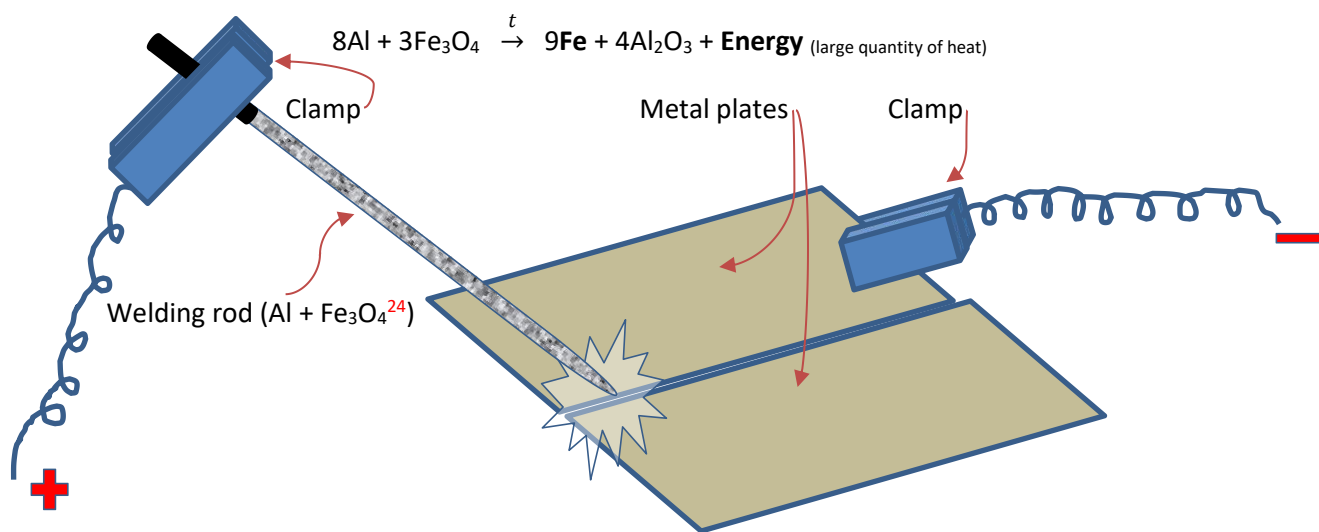
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Aluminium salts hydrolyze. Some are completely decomposed by water (e.g., Al_2S_3 and $\text{Al}_2(\text{CO}_3)_3$):



Welding mixture, **Al + Fe₃O₄**, when heated, produces a large quantity of energy (in form of heat), in this case, the produced iron melts (diagram):



²⁴When there is inadequate oxygen, iron forms mixed oxide, Fe_3O_4 ($\text{Fe}_2\text{O}_3 \cdot \text{FeO}$)

Aluminium is widely used as a construction material, and also in electro-industry (conducts heat well, better conductor than Cu).

Aluminium sulphate, $\text{Al}_2(\text{SO}_4)_3 \cdot 18\text{H}_2\text{O}$, is used for cleaning water and preparing some brand (variety) of paper.

Potash alum, $\text{KAl}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$, is used for skin tanning²⁵ (hardening) and also in paint works as a dye for cotton fabric.

²⁵Tanning – making leather from rawhide.

Aluminium is made passive by HNO_3 , therefore, it is suitable for handling this corrosive liquid.

Aluminium foil is used for wrapping chocolates and making bottle tops.

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Gallium, Indium, Thallium and their compounds

These elements belong to the rare ones and in any amount, significant concentrations in nature are not found. They are mostly obtained from zinc concentrates after smelting²⁶ of zinc from the concentrates.

²⁶Smelting – extracting metal by heating.

In a free state, these elements are silver-white soft metals with low melting points.

In air they are stable (thin oxide film), they do not decompose in water, but easily dissolve in acids, Gallium and Indium dissolve in alkalis.

Besides having a maximum oxidation state of, +3, they can display lower oxidation state. For Thallium, there are compounds in which it displays oxidation state of, +1.

Oxides and hydroxides of gallium(III) and indium(III) are amphoteric; thallium hydroxide, $\text{Tl}(\text{OH})_3$, has only basic properties.

Compounds of thallium(I) resemble, on one side, with compounds of alkali metals, on the other – with compounds of silver. Thallium(I) oxide, Tl_2O , energetically combines with water, forming hydroxide, TlOH , which is a strong base and also very soluble in water.

Majority of salts of thallium(I) easily dissolve in water, but salts of hydrohalic acids, TlCl and TlBr , are similar to salts of silver, AgCl and AgBr , and almost insoluble and distinctively photosensitive; exception is TlF , which like AgF , is very soluble in water.

Metallic gallium is used for filling quartz thermometers, which are used for measuring high temperatures. Gallium melts at 29.8°C , and boils only at 2205°C , so, such thermometers can measure up to 1000°C and higher. Alloys of Ga and Au are used in jewelry and dental prosthetic²⁷ works.

²⁷Prosthetic – branch of medicine dealing with production and use of artificial body parts.

Indium is used in place of silver for covering reflectors; reflectors covered with indium do not tarnish (dim) with time and their coefficient of reflection remains constant.

Halides of thallium are good transmitters of infrared rays. Therefore, they are used in optical apparatus which work in IR region of the spectrum. Thallium carbonate is used for preparing glass with high refractive capability. Thallium and its compounds are very toxic.

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Group IA elements

Elements of group IA are called **Alkali metals** and have electronic formula ns^1 . They are strong reducers. Their characteristic oxidation state is, +1. In this oxidation state, atoms of alkali metals are stable and are reduced with great difficulty.

A hydroxide is a compound which contains the hydroxo group, OH, e.g., H_2SO_4 and NaOH. A hydroxide of a nonmetal is an acid and a hydroxide of a metal (sometimes NH_4^+) is a base.

A base consists of metal (sometimes NH_4^+) and monovalent hydroxo group OH. The number of OH is determined by the valency of the metal. The most important chemical property of bases is their ability to form salt with acids.

Bases are classified according to their solubility in water and according to their strength. On solubility, bases are divided into soluble or alkalis and into non soluble. The most important alkalis – NaOH, KOH and $Ca(OH)_2$.

On strength, bases are divided into strong and weak. Strong bases are all alkalis except ammonium hydroxide.

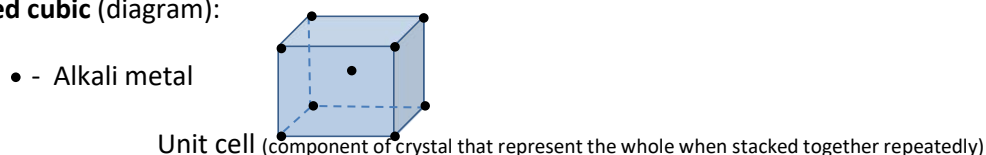
In nature, alkali metals are found in form of compounds (salts): chlorides, sulphates, carbonates, silicates and so on.

Because of the stability of compounds of alkali metals at oxidation state, +1, these metals can be obtained only by electrolysis of their salts' fusion or by action of stronger reducers. Example, sodium is produced by electrolysis of NaCl fusion and potassium, by passing sodium vapour through potassium

chloride:

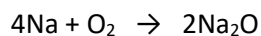
$$KCl + Na \xrightarrow{800^\circ C} K + NaCl$$

Alkali metals are soft, they are easily cut by a knife, on a freshly cut surface they appear silver in colour. They are all light and very fusible metals with good electro conductivity. In vapour state, atoms of alkali metals form molecules E_2 , e.g., Na_2 . The characteristic lattice for alkali metals in the crystalline state is **body centred cubic** (diagram):



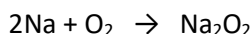
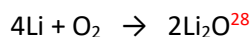
Chemical properties of alkali metals

Alkali metals are very active. Their activity increases in the series from Lithium to Francium, with an increase in their radii and fall in ionization energies. Air oxygen oxidizes them at normal temperature:

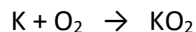


And that is why they are stored under a layer of hydrocarbon (kerosene).

On heating, alkali metals form oxide Li_2O , peroxide Na_2O_2 or superoxide KO_2 .



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²⁸Only lithium on burning forms normal oxide, Li_2O . Lithium ion by virtue of its small size, is not able to surround itself by sufficient peroxide ions to give a stable crystal lattice, hence, only the monoxide Li_2O exists.

Other alkali metals transform to peroxides Na_2O_2 , KO_2 , RbO_2 and CsO_2 .

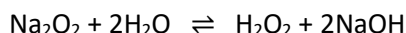
Na_2O can be obtained by passing moderate quantities of O_2 over sodium that is heated at temperature not more than 180°C or by heating sodium peroxide with metallic sodium:



Na_2O reacts vigorously with water with the formation of NaOH and release of a large quantity of heat:

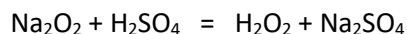


Na_2O_2 interacts with water to give H_2O_2 and NaOH :



$\text{O}_2^{2-} + 2\text{H}^+ \rightleftharpoons \text{H}_2\text{O}_2 + 2\text{OH}^-$, if the obtained solution is heated, then, as a result of decomposition of H_2O_2 , atomic oxygen is released from it.

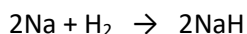
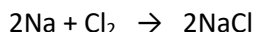
When dilute acid is acted on sodium peroxide, hydrogen peroxide is obtained:



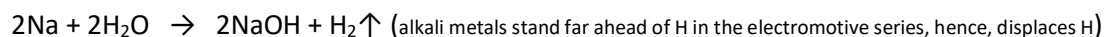
The process of interaction between Na_2O_2 and CO_2 has an important meaning:

$2\text{Na}_2\text{O}_2 + 2\text{CO}_2 = 2\text{Na}_2\text{CO}_3 + \text{O}_2$ This reaction gives basis for use of sodium peroxide in generating air in confined spaces, e.g., in a submarine.

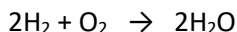
Alkali metals actively react with nonmetals (halogen, hydrogen, Sulphur and phosphorous):



Alkali metals react with water, forming the corresponding hydroxide and hydrogen:



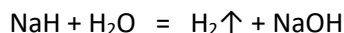
Since the reaction is exothermic, hydrogen ignites:



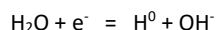
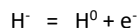
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Chemical properties of compounds of alkali metals

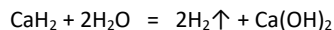
Hydrides of alkali metals are salt-forming substances, e.g., NaH^{29} reacts with water:



²⁹Compounds of hydrogen with metals are called **hydrides**. Hydrides of alkali and alkaline-earth metals are salts, i.e., the chemical bond between hydrogen and the metal in them is ionic. They are white crystals. They are all unstable and, on heating, decompose to metal and hydrogen. When acted upon by water, redox reaction takes place in which hydride ion, H^- , acts as a reducer, and hydrogen of water – as an oxidizer:

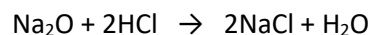
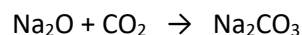


As a result of the reaction, hydrogen and base are produced:

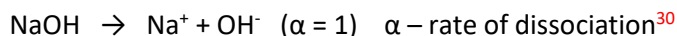


This reaction is used for determining traces of vapour and for their removal.

Oxides of alkali metals are active basic oxides:



Hydroxides of alkali metals are soluble hydroxides – alkalis. In this case, the rate of dissociation increases from LiOH to CsOH .



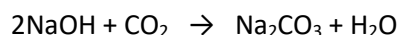
³⁰Rate of dissociation of an electrolyte is the ratio of the number of its molecules that have disintegrated in solution into ions to the total number of its molecules in solution.

NaOH is solid white hygroscopic crystals, with melting point, 320°C . Because of its strong corroding action on clothes, skin, paper and other organic substances, it is called **caustic sodium**. In industry, NaOH is called **caustic soda**.

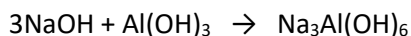
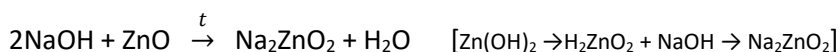
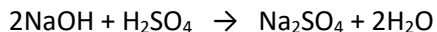
NaOH should be stored in well-sealed containers, since it easily absorbs CO_2 from air and slowly changes into sodium carbonate.

Na^+ ion attracts 25 water molecules, hence, the more a person consumes salt, the more water will be in the blood vessel and the higher the blood pressure will rise.

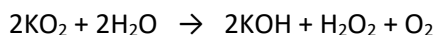
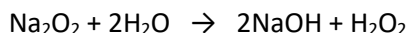
Other reactions:



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Peroxide Na_2O_2 and superoxide KO_2 are strong oxidizers, they are decomposed by water, forming H_2O_2 and O_2 :



Salts of alkali metals dissolve well in water (except salts of lithium³¹), they colour bunsen burner flame red (lithium salts), yellow (sodium salts), light violet (potassium salts). Salts of alkali metals with weak acids hydrolyze, forming an alkaline media, e.g., NaCN .

³¹All alkali metals are extremely electropositive, increasingly so with increasing atomic number; they therefore, tend to form predominantly electrovalent compounds and these are generally soluble in water. Lithium, because of the small size of its atom and unipositive ion, tends to form compounds in which there is some degree of covalent character – and behaves like Magnesium.

The appreciable covalent character of several compounds of Lithium depends on the fact that its lone valency electron is not completely transferred during chemical reactions – because of the small size of lithium atom.

NaCl – common salt. A person uses 5 – 10 kg in a year.

NaNO_3 – nitrogen fertilizer.

Soda, Na_2CO_3 , is used in the production of glass, soap, in textile and paper industry.

Excess of sodium salts in the soil (salinization of soil) erases their fertility.

Salts of potassium KCl , KNO_3 and KPO_3 are fertilizers.

The external form and also the physical and chemical properties of potassium very much resemble those of sodium, but possess even bigger activity.

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Group IIA elements

Group IIA elements have electronic formula ns^2 . In group IIA are Beryllium, Magnesium, Calcium, Strontium, Barium and Radium. All these elements except Beryllium, clearly display metallic properties.

In a free state, they are silver-white substances, more hard than alkali metals with high melting points. In density, they are all (except Radium) light metals.

These elements are strong reducers, less active than alkali metals. Their characteristic oxidation state is, +2, and valency of 2. During formation of a covalent bond, $s \rightarrow p$ electron excitement and sp – hybridization of AOs takes place.

Elements of group IIA can be divided into alkaline-earth³² metals: Ca, Sr, Ba, Ra, whose bases are alkalis, and Be, Mg, whose bases are less soluble in water.

Beryllium in its properties is nearer to Aluminium (diagonal resemblance).

³²The source of the name “alkali-earth” is connected to the fact that hydroxides of Ca, Sr and Ba, just like hydroxides of potassium and sodium have alkali properties, but the oxides of these metals in their fusibility are similar to oxides of Aluminium and heavy metals, which were earlier commonly called earth. Aluminium found in clay plus other rare-earth elements are given the name earth metals.

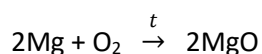
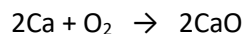
In nature, they are found in the form of compounds of salts: sulphates, carbonates, phosphates and silicates. These elements are produced by electrolysis of their salts' fusions. For Be – beryl $\text{Be}_3\text{Al}_2(\text{SiO}_3)_6$, for Mg – magnesite MgCO_3 and dolomite $\text{MgCO}_3 \cdot \text{CaCO}_3$ ($\text{MgCa}(\text{CO}_3)_2$) and for Ca – lime, chalk and marble.

Chemical properties of the elements

Elements of group IIA are less active reducers than are alkali metals. Their reducing properties increase from Beryllium to Radium.

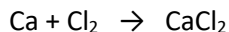
Air oxygen oxidizes Ca, Sr, Ba and Ra at normal temperature (just like alkali metals).

Mg and Be are covered by an oxide film and oxidizes only when heated. They don't corrode and also almost don't react with water.



Being active reducers, metals of group IIA react with nonmetals:

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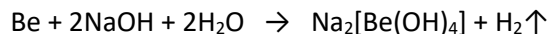


With water: $\text{Ca} + 2\text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2 + \text{H}_2\uparrow$

With acids: $\text{Mg} + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{H}_2\uparrow$



For Beryllium, it dissolves in aqueous solutions of alkalis and forms hydroxoberyllates:



Magnesium is obtained through electrolysis. Molten magnesium chloride is subjected to electrolysis.

Magnesium is a very light silver-white metal.

The main area of use for metallic magnesium is for obtaining different light alloys. Adding other metals to Magnesium gives the alloys strength, hardness and resistance to corrosion. The most valued properties are in alloys called **electrons**. These relate to three systems: Mg – Al – Zn, Mg – Mn and Mg – Zn – Zr. The most widely used is Mg – Al – Zn. These alloys are used in rocket technology and aviation industry due to their lightness.

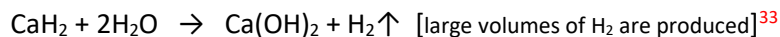
The main use of Beryllium is in alloys – where it acts as a regulating additive.

Beryllium is lighter than Aluminium but stiffer than steel.

BeO is a white powder that is very toxic when inhaled because it affects the lungs.

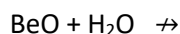
Chemical properties of the compounds

Hydrides of alkaline-earth metals are ionic compounds and interacts with water:

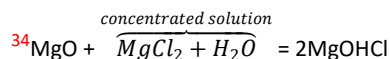
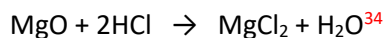
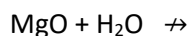
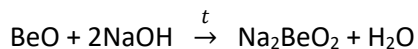
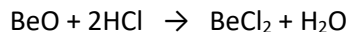


³³Because of its ability to produce a large volume of H₂, CaH₂ is sometimes used for obtaining this gas. When 1 kg of CaH₂ is interacted with water, almost 1000 litres of H₂ are produced. CaH₂ is used as an effective drier, able to remove water even from crystal hydrates.

Oxides of alkaline-earth metals Ca, Sr, Ba and Ra dissolve in water with the formation of alkalis, oxide of Mg is less soluble in water and has basic properties, oxide of Be, which is insoluble in water, has amphoteric properties:



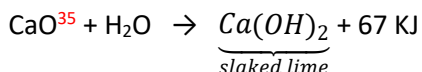
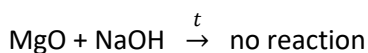
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MgOHCl has important industrial value. The technical product is obtained by mixing MgO with concentrated aqueous solution of MgCl_2 and is called **Magnesian cement** (Sorel's cement). This mixture after sometime hardens (solidifies), changing into a thick white polishable mass. Solidification can be explained by the fact that the hydroxochloride, MgOHCl, polymerizes into chains, $-\text{Mg}-\text{O}-\text{Mg}-\text{O}-\text{Mg}-\text{O}-\text{Mg}-$, at the ends are Cl atoms or hydroxyl groups OH.

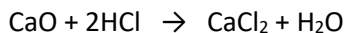
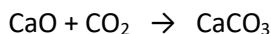
Magnesian cement as a binding material is used when preparing milling or grinding stone and different slabs or tiles or plates.

Natural silicates of magnesium, especially Asbestos, $\text{CaO} \cdot 3\text{MgO} \cdot 4\text{SiO}_2$, are used as heat insulators.

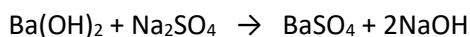
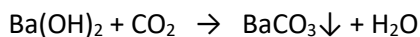
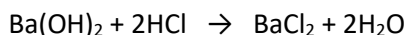
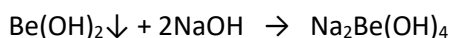
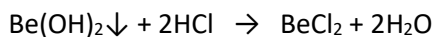


³⁵CaO is a white substance. It is a strong fire resistant substance and melts only at 3000°C . It is called **unslaked (quick) lime**.

If quicklime is doused (drenched) with water, then water is absorbed by porous (spongy) pieces/chunks of lime and reacts with it releasing a significant quantity of heat (equation above). As a result, part of the water turns into vapour, and pieces of lime scatter (crumble) into loose mass of calcium hydroxide. This operation is called **lime slaking**, and the obtained product is called **slaked lime**.

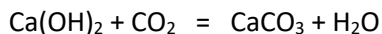


Hydroxides of Ca, Sr, Ba and Ra are alkalis, hydroxide of Mg – less soluble basic hydroxide, hydroxide of Be – amphoteric hydroxide.



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Slaked lime is used in building works. Its mixture with sand and water is called **lime solution** and is for strengthening bricks when building a wall. Slaked lime is used also as a **plaster**. Hardening of lime first takes place because of evaporation of water, and then as a result of the slaked lime absorbing carbon monoxide from the air, and formation of CaCO_3 :



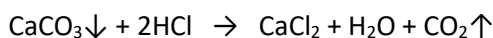
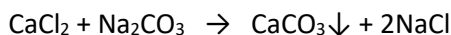
Because of the small amount of CO_2 in air, the process of hardening is slow, and since in this case water is evolved, then, in buildings, built using lime solution, dampness is retained for long.

*Normal cement (silicate cement) is obtained by roasting a mixture of clay with slaked lime.

Chemical composition of cement: CaO , Al_2O_3 , SiO_2 and Fe_2O_3 .

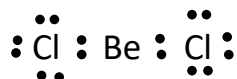
Carbonates and sulphates of elements of group IIA are less soluble in water.

Carbonates dissolve in acids:



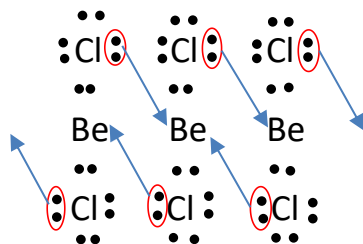
In forming compounds like BeX_2 , e.g., BeCl_2 , sp -hybridization takes place (linear structure).

The common number of valence electrons in the molecule BeCl_2 is insufficient to fully fill the outer electronic level of Beryllium atom. Hence, this type of molecule is called **electron-deficient**.



In the outer level of Beryllium atom, there are only four electrons. Therefore, Beryllium atom is able to be an acceptor of electron pairs and form two more covalent bonds via donor-acceptor.

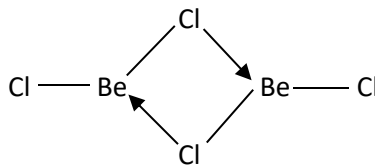
At the same time, each chlorine atom in the molecule BeCl_2 has unshared pairs of electrons and can be donors. Therefore, on cooling (condensing) gaseous BeCl_2 , new covalent bonds appear between separate BeCl_2 molecules:



Solid form $(\text{BeCl}_2)_n$

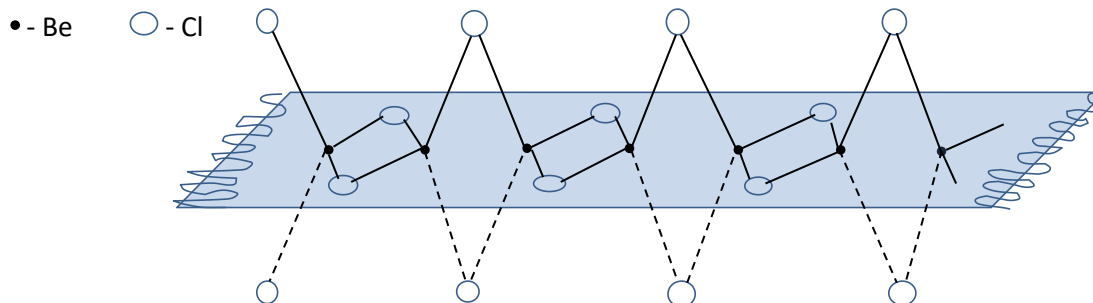
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In gas form, BeCl_2 forms a **dimer**:



Gas form $(\text{BeCl}_2)_2$

To sum up, linear polymeric chains are formed during the condensation of BeCl_2 , in which chlorine atoms play the role of bridges that bond Beryllium atoms. Chlorine, here, is a bridging atom (diagram, below). From the diagram, it is evident that the covalency and coordination number of Beryllium in solid BeCl_2 is four.



Polymer $(\text{BeCl}_2)_n$

This covalency and coordination number is characteristic of many stable compounds of Beryllium (compare with $(\text{Al}_2\text{Cl}_3)_2$).

Because of Calcium being so widespread, its salts are almost always found in natural water. If water contains CO_2 , the calcium carbonate can enter into solution in form of hydrocarbonate $\text{Ca}(\text{HCO}_3)_2$. Natural water containing in solution a big amount of salts of Ca and Mg is called **hard water**.

Soft water can have a small amount of salts of Ca and Mg or none at all.

Ca^{2+} and Mg^{2+} ion content is responsible for hardness of water: **temporary**, if there are hydrocarbonates of Ca and Mg, and **permanent**, if in water there are chlorides and sulphates of Ca and Mg.

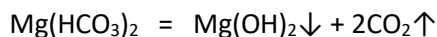
On heating hard water, precipitation takes place as a result of decomposition of calcium hydrocarbonate:



Softening of hard water takes place as a result of decomposition.

On heating hydrocarbonates of Mg, precipitation takes place as a result of decomposition of Magnesium hydrocarbonate:

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As a result, water is softened.

Strontium. Barium

They are found in nature in form of sulphates and carbonates.

Metallic Strontium and Barium are very active, quickly oxidizes in air, energetically reacts with water (especially Ba) and directly combines with many other elements.

Oxides of Strontium and Barium, SrO and BaO resemble calcium oxide. Both metals form peroxides. Barium peroxide, BaO₂ is obtained by heating BaO in air at 500 °C. At higher temperatures, it decomposes to oxide and oxygen.

BaO₂, just like Na₂O₂, is used for whitening materials. BaO₂ interacts with water to give H₂O₂ and the hydroxide. H₂O₂ easily decomposes to give atomic oxygen and water. Atomic oxygen is the reason for its ability to whiten materials.

Hydroxides of Strontium and Barium, Sr(OH)₂ and Ba(OH)₂, are strong bases, they are more soluble in water than is Ca(OH)₂. Concentrated solution of Ba(OH)₂ is called **Baryta water** and is frequently used as a reagent.

Salts of Strontium and Barium have similarities to salts of calcium. Carbonates and sulphates SrCO₃, BaCO₃, SrSO₄ and BaSO₄ have very little solubility in water and precipitates from solution.

The characteristic difference of all three metals (Ca, Sr and Ba) from each other is the colour conveyed by their volatile salts' nonluminous flames. Calcium salts colour the flame brick-red, salts of Strontium – crimson red, salts of Barium – yellowish-green.

Salts of Barium, BaCl₂ and BaCO₃, are poisonous and are used as insecticides.

Magnesium is a chief construction material, it is a microelement.

Ca(OH)₂ is used in building.

Salts of Calcium, for example, gypsum CaSO₄.2H₂O is used for gypsum application of saline soils.

Exercise: By what type of hybridization of Beryllium's AOs is the formation of:

- i) The molecule BeCl₂
- ii) The dimer (BeCl₂)₂
- iii) The polymer (BeCl₂)_n?

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Write the reaction equations that demonstrate the amphoteric properties of BeO and Be(OH)₂.

(Use rubidium hydroxide and hydrochloric acid)

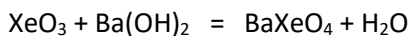
Using an energy level diagram, describe the bonding in the bridging group Be – Cl – Be of the dimer (BeCl₂)₂.

Group VIIIA elements (The noble gases)

Group VIIIA elements have electronic formula ns² for Helium and ns²np⁶ for Neon, Argon, Krypton, Xenon and Radon. These elements are characterized by very low chemical activity, which gave them the name **noble** or **inert gases**. Only with difficulty do they form compounds with other elements or substances; chemical compounds of Helium, Neon and Argon have not been obtained.

Atoms of noble gases are not combined into molecules, in other words, their molecules are **monoatomic**.

Krypton, Xenon and Radon are able to form compounds with other elements and especially with Fluorine. Direct interaction between the noble gases and Fluorine (on heating or electric charge) has produced fluorides KrF₂, XeF₂ (linear), KrF₄, XeF₄ (square planar) and RnF₄. All are crystals and are stable under normal conditions. Derivative of Xenon at oxidation state of, +6, has been obtained – hexafluoride XeF₆ (pentagonal bipyramid), trioxide XeO₃ (tetrahedral), hydroxide Xe(OH)₆. The last two compounds display acidic properties; so, when they react with alkalis, they produce salts of xenonic acid, H₂XeO₄, for example:



Derivatives of Xenon(IV) are strong oxidizers. However, when stronger oxidizers act on them, compounds in which Xenon is at oxidation state, +8, are obtained. Examples are octafluoride XeF₈, tetraoxide XeO₄ and hexafluorooxide XeOF₆.

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The higher chemical activity of Krypton, Xenon and Radon in comparison to the first members of the group of noble gases is explained by the relative low potential of ionization of their atoms.



In Neon, the valence electron is nearer to the nucleus

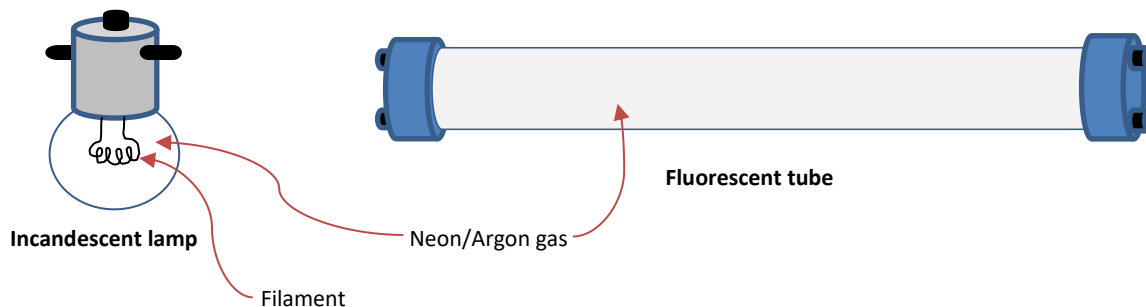
In Xenon, the valence electron is further away from the nucleus

Therefore, a very electronegative element like Fluorine is able to pull away an electron from Xenon and form a compound with Xenon.

More practical use is found for Argon, Neon and Helium.

Helium is widely used in physical laboratories as a cold carrier. It acts also as a thermometric substance in thermometers that work at temperature intervals of 1 to 80 K. Isotope of Helium, ^3He , is the only substance suitable for measuring temperatures lower than 1 K.

Neon and Argon are widely used. They are used for filling incandescent lamps. They are used for filling fluorescent tubes; for Neon – red lighting, for Argon – bluish lighting.



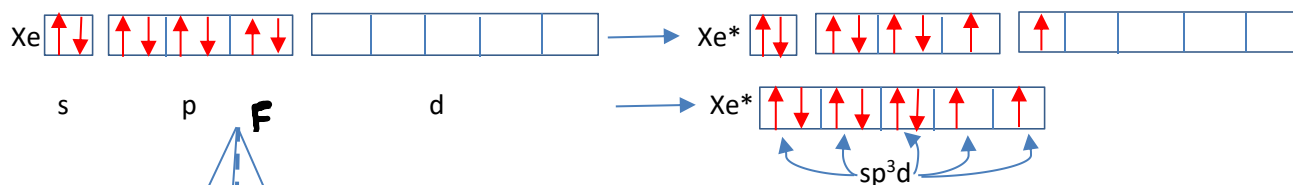
Argon, being the most available, is used in metallurgy and chemical processes that requires inert media.

The structures of Xenon difluoride, XeF_2 , and Xenon tetrafluoride, XeF_4 , have been proven to be linear and square planar, respectively; there is also evidence that Xenon hexafluoride, XeF_6 , assumes a non-octahedral configuration, and Xenon trioxide, XeO_3 , assumes a tetrahedral configuration.

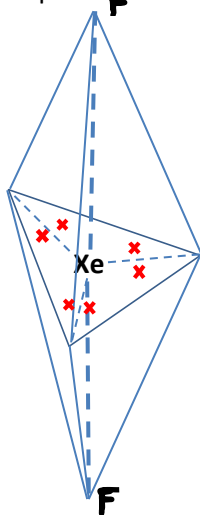
These structures can be explained by the simple theory of electron pair repulsion.

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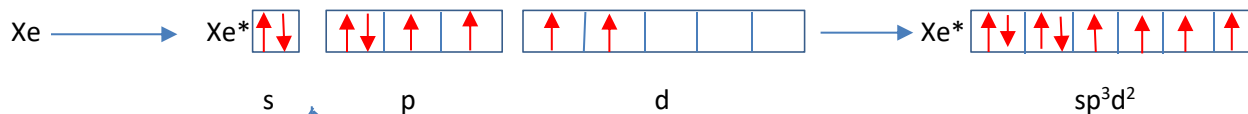
The bonding in these compounds is presumably predominantly covalent and involves electron promotion from the p- sublevel to the unoccupied d- sublevel in the Xenon atom. Fluorine atom, being extremely electronegative, will bring this about.



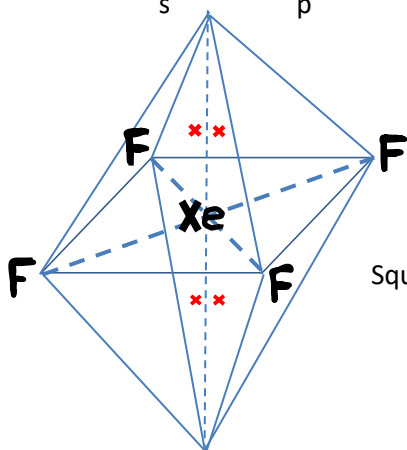
XeF_2



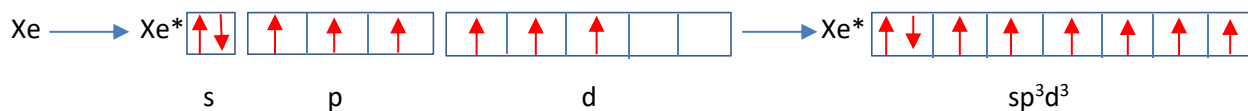
Linear structure (from electron diffraction + spectroscopic methods)



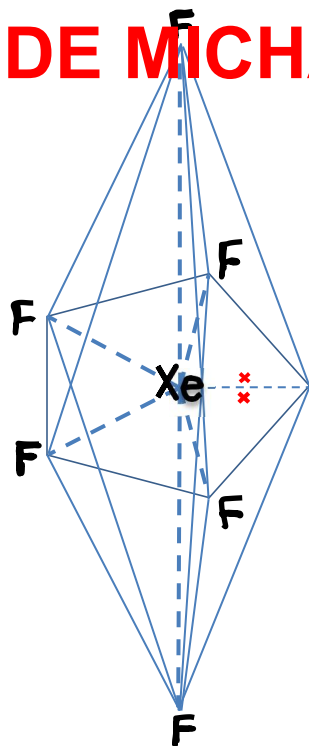
XeF_4



Square planar (electron diffraction + spectroscopic methods)

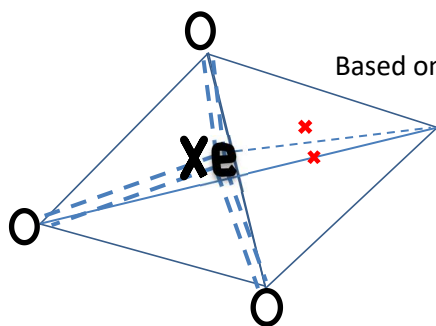
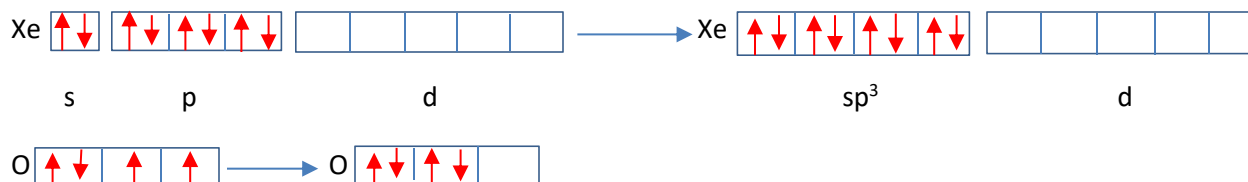


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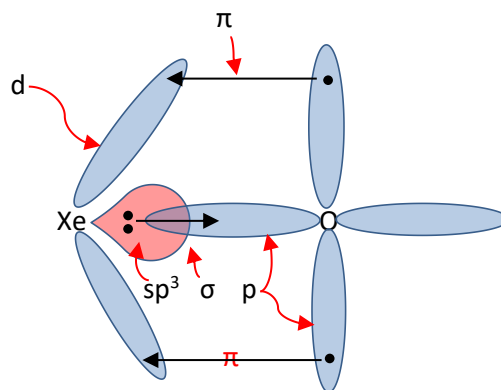


Based on pentagonal bipyramid (non-octahedral)

[IR + electron diffraction methods]



Based on tetrahedron



When the noble gases Argon, Krypton and Xenon are compressed with water, **hydrates** are formed containing six water molecules for each noble gas atom, e.g., $\text{Xe} \cdot 6\text{H}_2\text{O}$.

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These compounds are in reality enclosure compounds (clathrates³⁶) in which the noble gas atoms are trapped in a network of water molecules, the water molecules being held together by hydrogen bonding.

Similar types of clathrates are formed when noble gases at pressures of 10 – 40 atmospheres are brought in contact with a solution of quinol, $C_6H_4(OH)_2$, in water. On crystallization, crystals, different from those of pure quinol are formed; they contain noble gas atoms trapped in a quinol network and are quite stable. On heating, the hydrogen bonds holding individual quinol molecules together are broken, and the noble gas is released.

³⁶It must not be thought that clathrates are true chemical compounds, since there is no evidence to suggest that chemical bonds involving the noble gases are present.

Lecture notes compiled and written by Ben Ogola, PhD, Chemistry (Moscow)